

Rule-Rewriting Cellular Automata and the Edge of Chaos

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Abstract

We study a two-layer cellular automaton in which each cell carries both a phenotype state and an eight-bit rule. A permutation σ of the eight displayed truth-table positions defines an operator P_σ on those rules by $\text{bit}[\sigma(p)] := \neg \text{bit}[p]$. The resulting family of $8! = 40,320$ permutation-propagators admits exact classification rather than sampling: P_σ has a fixed rule exactly when every cycle of σ is even, an all-even permutation with k cycles has 2^k fixed rules, and every fixed rule is balanced, with Langton activity $\lambda = 1/2$.

This places every fixed point of every operator at exactly the value the classical picture associates with the transition between chaos and order. However, in this setting, λ is powerless to distinguish an operator whose rule field stays diverse from one that collapses. A census across all cycle types separates sustaining from collapsing operators where λ cannot, and a finite-size scan over $L = 30\text{--}240$ finds a broad, drifting crossover rather than a sharpening critical point. The classical search, conducted in parameter space, returns nothing.

We then leave the permutation core while retaining the same two-layer genotype–phenotype substrate. The objects are now update *architectures* on this substrate, constructed from simple operators including those mentioned above, along with temporal schedules, phenotype-reading constructions, mutation, and conservative transport. We use the reach of damage-spreading to characterise these architectures, calibrated against rules

of undisputed regime. Sustained rule diversity is an analogue of the order parameter from the classical account, however, we develop a conservative control that holds the rule histogram invariant while varying transport, and show that damage-spreading can be effectively unchanged with regard to diversity while moving by an order of magnitude with transport. Within the tested architectures, differences in causal reach are driven by genotype-to-phenotype *coupling*. That coupling is a gain rather than a switch: as the genotype’s view of the phenotype moves from an ancestral copy to the causally current one—marginal statistics matched throughout, so that only currency varies—reach rises monotonically out of the ordered band and into the complex one fixed by the elementary rules. How legible acquired state is to the heritable layer is therefore a quantitative control on dynamical regime rather than a categorical distinction. We then make that gain endogenous, replacing the dial with a local predicate. A predicate blind to the phenotype reproduces the dial exactly, linear in its firing fraction; one that reads the phenotype exceeds it, by margins growing with the width of the neighbourhood read; and an otherwise identical predicate reading a phenotype frozen at the perturbation time does not exceed it at all. Since width is the spatial correlation length of the firing pattern, correlated firing alone accounts for none of the gain: what a self-setting coupling exploits is that what it reads is current.

Keywords: cellular automata, rule evolution, Langton’s lambda, self-modifying systems, edge of chaos, artificial life, permutation operators, exhaustive classification, damage spreading, perturbation analysis

Two Readings of the Edge of Chaos

In 2015, working on a different problem, we built a cellular automaton whose cells carried rules rather than states (Corneli & Maclean, 2015). A programming error connected with one example generated spacetime diagrams reminiscent of the edge of chaos behaviour observed in classical CAs. The question of how that example recovered the distinctive visual behaviour remained open.

Recent computational experiments have located that accidentally discovered operator within a larger landscape. A permutation σ of the eight truth-table positions of an elementary rule defines an operator P_σ that rewrites a rule by negating the bits σ names; there are $8! = 40,320$ such operators, and they admit exact classification rather than sampling. They form the bijective core of the 8^8 possible writings. The original example recreated in Figure 3 is a non-bijective writing outside that core. We therefore separate two objects that share a substrate but not a scope: the finite permutation core studied in Part I, and the wider class of state-dependent update architectures studied in Part II.

The classical literature offers two readings of “the edge of chaos”, and they are different kinds of object. The first is a location in *parameter space*, aligned with an order parameter which can be tuned so that the dynamics sit between order and disorder (Langton, 1990). Our classification lands on it from an unexpected direction—every fixed rule of every operator in the family is balanced, with Langton activity $\lambda = 1/2$, exactly the value the classical picture associates with the transition. Yet searching the Meta Cellular Automaton (MetaCA) for a critical transition does not produce one (Supplement 1, Box 6). The algebra places the fixed points precisely at the classical critical value; the dynamics show a broad, drifting crossover and no critical point of that sort at all.

The second reading is not a boundary one tunes toward but a region one measures the system into. It is dynamical: an ordered system absorbs a perturbation, a chaotic one lets it fill the light cone, and the interesting regime lies between. That reading requires no parameter sweep and no choice of where to draw a region—it requires only a perturbation

and a clock. Flipping a single cell and tracking the divergence against an otherwise identical run, on a scale calibrated against elementary rules whose regime is not in dispute, places the phenotype-reading construction of *Constructive control by temporal schedules* between rules 90 and 54, below rule 110 and far below the chaotic rule 30, while frozen rules return exactly zero (*A Causal Measure*). The live phenotype-to-genotype coupling is what puts it there: cutting that single edge, with seed, tape and construction otherwise identical, roughly halves how far a perturbation reaches.

Pursued far enough, the second reading also tests and rejects a tempting substitute for the classical order parameter. Across seven mechanisms, causal reach initially appears to rise and then fall with sustained genotype diversity, but the descending observations obtain high diversity only through mutation or by holding rules still. A conservative eighth construction prescribes the genotype measure and transports it by local bijective exchanges. It sustains causal amplification across the full 32–256 rule range: exchange rate, not diversity alone, controls reach (*Diversity and conservative transport*).

Placing the tested architectures on the calibrated scale then identifies a candidate control within the extension. The constructions separate by *coupling*: where the genotype update never reads the phenotype—the bare operators, braids, niches, refuges, mutation, and blend strength across its whole range—nothing clears the bottom of the complex band, while everything that reads it lands inside (Supplement 1, Box 10). The coupling is moreover a dial rather than a switch. Tuning the gain of that loop carries reach smoothly from the frozen identity past rules 90, 110 and 54, with a locatable crossing; an ungated control matched in mobility, differing only in that its rewriting is blind to the phenotype, never leaves the ordered band. Within these constructions the classical picture is therefore recovered in a coordinate it does not usually use: not a truth-table parameter, but the gain of the loop from the phenotype back to the genotype.

This is the paper’s edge-of-chaos claim, and its scope is the feedback-coupled extension, not the 8! family as a whole. It is measured rather than assumed: in the tested

construction, adding the phenotype-to-genotype edge carries causal reach into the calibrated complex band and cutting it carries reach back.

Supplement 4 reports a separate exploratory analysis of coexisting spatial domains and documents a boundary-transport statistic withdrawn after a circularity check. Neither is used to support either edge-of-chaos claim here.

Part I classifies the permutation-propagators, surveys their spatial dynamics, and tests the parameter-space reading of the edge of chaos. Part II explicitly drops the restriction to a single permutation-propagator and studies causal reach in update architectures built on the same genotype–phenotype substrate. Part III makes the gain of that coupling endogenous rather than imposed, and asks whether a composition whose condition reads the lattice escapes the coordinate Part II identifies. The exact results hold for the entire $8!$ core. The causal measurement covers one principal construction at one lattice width over six seeds, and no comparable causal census of either the $8!$ or 8^8 family is attempted here. Figure 1 maps that division of labour and marks where the scope boundary falls.

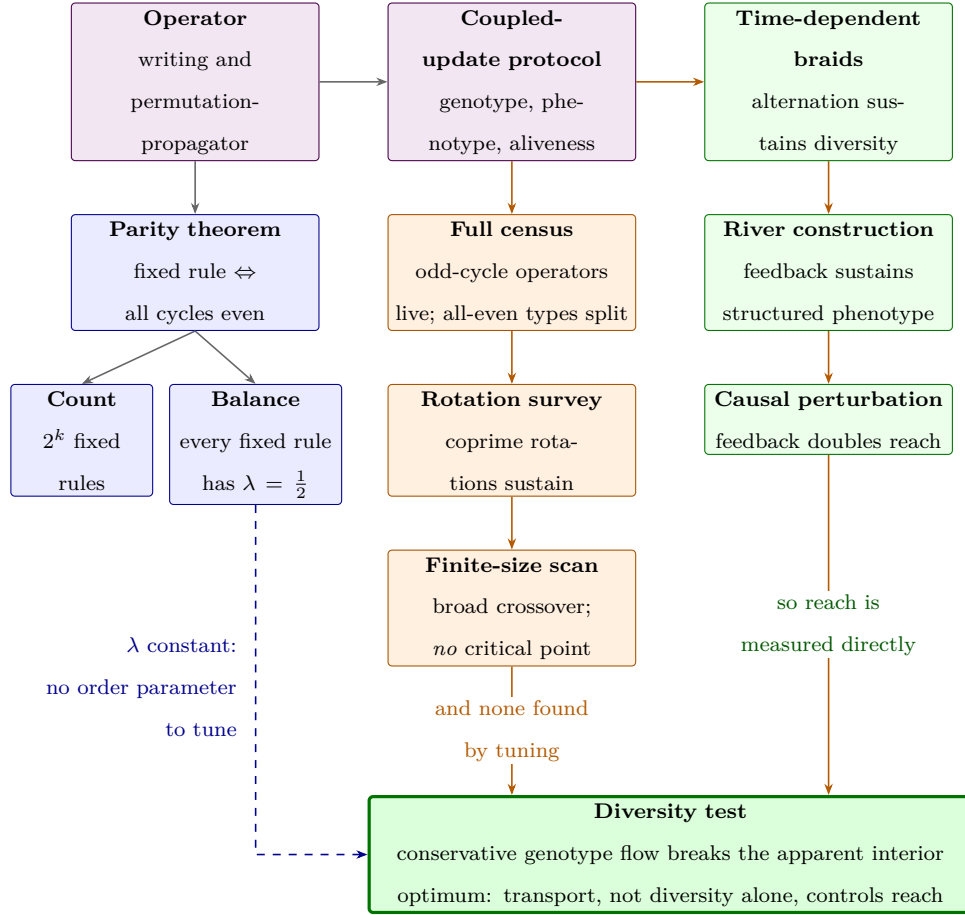


Figure 1

Argument map and scope boundary. Part I contains the exact branch (blue) and the empirical parameter-space branch (amber) inside the permutation core. The constant value $\lambda = 1/2$ closes off that classical coordinate there; it motivates looking elsewhere but does not entail what follows. Part II is the constructive branch (green): it leaves the single-operator family, introduces time- and phenotype-dependent update architectures, and measures causal propagation. Arrows mark dependence; the change of object between the amber and green branches is explicit.

Part I

The Permutation Core

Definitions, theorems, corollaries and worked examples are stated and proved in Supplement 2, and referred to here by number. The narrative gives each result in prose at the point where it is used, so this part can be read without turning to the supplement; the supplement carries the formal statements, the proofs, and the arithmetic worked by hand.

The 2015 Example

The operator classified in Part I was reached by working out what a programming error actually did. That derivation is short, it accounts for the figure that motivated the enquiry, and it fixes the scope of everything that follows, so we give it first.

The 2015 engine mutated a rule by reading the output at one displayed position and writing its negation one position to the left, clamping at the end:

$$\text{bit}[\max(k-1, 0)] := \neg \text{bit}[k].$$

Here k indexes that engine's own display order, which is not the Wolfram order adopted from *The Object* onward. The distinction does not affect the argument of this section, which concerns only the shape of the map, but it does affect any numerical comparison carried across the two conventions.

Two consequences follow at once, and both are visible in the engine's write log. Position 7 is never written—nothing shifts into the top—while position 0 is written twice, once by $k = 0$ and once by $k = 1$. Over 87,632 recorded writes the measured shares are 0.000% and 24.9% $\approx 2/8$. A map that leaves one position without a preimage and gives another two is neither injective nor surjective. It is not a permutation, and it is therefore not a member of the $8!$ family classified below.

The same two properties account for the figure. Information cascades $7 \rightarrow 6 \rightarrow \dots \rightarrow 0$, negating at each step, so the configuration consistent with the cascade is

the alternating one, $g(k-1) = \neg g(k)$. Position 7 receives nothing from this write and is determined by the remainder of the update; positions 6 down to 1 then fill with alternation and freeze. Position 0 cannot settle at all, because $k=0$ demands $g(0) = \neg g(0)$, which no assignment satisfies. The field is therefore driven to exactly two configurations differing in exactly one position, 00101010 and 10101010. Read directly as eight-bit numbers these are the 42 and 170 reported with the original figure—the genotype string itself, not a conversion into standard rule numbering. The residual is a hunt on the single unsettleable position rather than a pair of fixed configurations.

That distinction sets the scope of what follows. The mechanism just described requires a position with no preimage and a position with two, and a permutation has neither; no member of the $8!$ family can reproduce it. Part I therefore studies a neighbouring object, reached from this one and sharing its elementary write, rather than a generalisation of it. The permutations are the bijective core of the 8^8 writings, and the 2015 example lies in the remainder.

The Object

A cellular automaton normally evolves a field of *states* under one fixed rule. Here each cell also carries a *rule*, and local writes evolve that rule field alongside the phenotype. We display an elementary cellular automaton (ECA) rule in Wolfram order,

$$111, 110, 101, 100, 011, 010, 001, 000,$$

and index these displayed positions from left to right by $p = 0, \dots, 7$. Thus position 0 is the 111 output and most significant bit, while position 7 is the 000 output and least significant bit. Throughout, the indices of σ , w , g , and $\text{bit}[p]$ are these *displayed positions*, not the binary values of the neighbourhoods. If $n \in \{0, \dots, 7\}$ is a numerical neighbourhood value, its rule output is therefore $g(7-n)$.

We study the operator P_σ , parameterised by a permutation $\sigma \in S_8$ of the displayed positions. Writing the displayed byte as a map $g: \{0, \dots, 7\} \rightarrow \{0, 1\}$, the operator sends g

to $P_\sigma g$ defined by

$$(P_\sigma g)(q) = \neg g(\sigma^{-1}(q)), \quad \text{equivalently} \quad \text{bit}[\sigma(p)] := \neg \text{bit}[p]. \quad (1)$$

Each output bit is one input bit, at the position σ moves it to, negated; P_σ is thus a permutation of the eight coordinates composed with a global complement. The truth table is thereby itself an eight-cell automaton whose wiring is σ and whose update is Boolean negation — an automaton inside the automaton. Equation (1) defines a family of $8! = 40,320$ operators that is finite, exhaustively enumerable, and, as we show below, exactly classified.

The legacy engine stored its table in the order 000, 001, 010, 100, 011, 101, 110, 111. Whenever a legacy writing is imported, the reproduction code conjugates it into the displayed Wolfram order above before we quote its permutation or apply names such as offset +2.¹

Worked example 1 works through one member of the family bit by bit.²

The simultaneous operator acts on one eight-bit rule; the spatial experiments place its elementary writes inside a two-layer automaton.³

The substrate—a cellular automaton whose cells hold rules rather than states—is not itself new (Corneli & Maclean, 2015); what we study is the *operator* of Equation (1), understood as an enumerated, exactly classified family. The example in that 2015 paper is a non-bijective writing (*The 2015 Example*) and therefore lies outside this family; what it shares with the family is the elementary write, not membership. We give the family here an essentially complete characterisation. The canonical symmetry group of ECA rule space has order four (reflection $\times$ complement, collapsing 256 rules to 88 classes) (Schaller &

¹ Supplement 2, Definition 1: Writings, elementary writes, and propagators. Supplement 2, Observation 1: Permutations form a small bijective core.

² Supplement 2, Worked example 1: A simultaneous propagator by hand.

³ Supplement 2, Definition 2: Genotype and phenotype. Supplement 2, Worked example 2: One site, one complete MetaCA step.

Svozil, 2025); it is not S_8 , and the negation-carrying, σ -parameterised family of Equation (1) does not appear in it. Rule-space permutation as a parameterised, enumerated family with a derived fixed-point structure does not appear in the literature we have surveyed; the nearest neighbours augment cellular automata with state-dependent feedback (Pavlic et al., 2014) rather than permuting the rule itself. Two established genres are adjacent but distinct: per-cell rules (non-uniform or ν -CA (Dennunzio et al., 2011)) and rules that change during evolution (rule-changing or programmable CA (Mori et al., 1998)). Non-uniform cellular automata are not, by themselves, the source of novelty here: when each site draws its rule from a finite set, the resulting system can be represented as a uniform cellular automaton on a product alphabet (Dennunzio et al., 2011). Our contribution instead concerns the permutation-propagator family and its fixed-point structure.

The Classification

Fixed points exist iff every cycle is even

P_σ has a fixed rule if and only if every cycle of σ has even length; since a one-cycle is odd, this already forces σ to be fixed-point-free.⁴

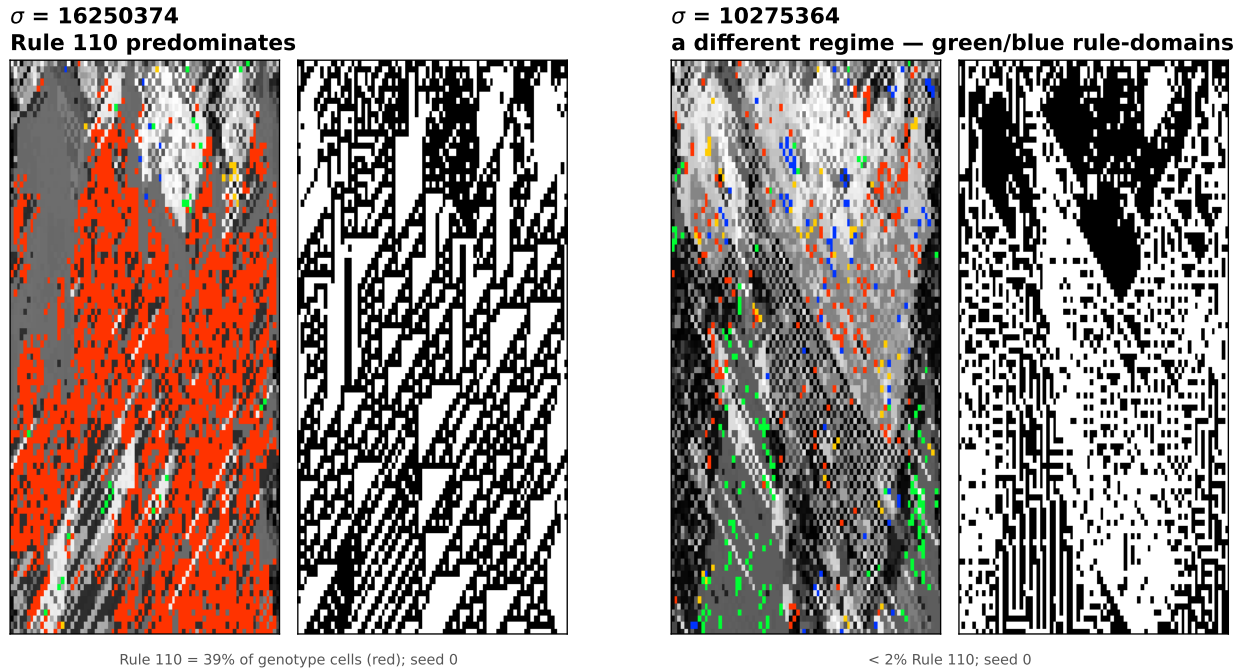
A fixed rule is a g invariant under (1), i.e. a σ -equivariant Boolean colouring with $g(\sigma k) = \neg g(k)$. Follow a cycle of length L : equivariance forces $g(k) = (\neg)^L g(k)$, and $(\neg)^L$ is the identity exactly when L is even; conversely, colour each orbit by the parity of its distance from a chosen representative. A one-cycle of σ ($L = 1$) is odd and so admits no such colouring, which is why even cycles are required throughout.

Worked example 3 shows the theorem on the offset-+2 operator. The same small calculation also makes the later 2^k count and forced $\lambda = 1/2$ balance visible before we state them generally.⁵

This is the signed-cycle fixed-point theorem of Boolean-network theory (Aracena,

⁴ Supplement 2, Theorem 1: Fixed points and even cycles.

⁵ Supplement 2, Worked example 3: Even and odd cycles.

**Figure 2**

Two operators from the family, each shown over time as genotype (256-colour, left) and phenotype (black and white, right). Under $\sigma = 16250374$ the genotype field comes to be dominated by ECA Rule 110—the red cells, 39% of the genotype at the representative seed (35% over six seeds)—and the phenotype accordingly displays Rule 110’s characteristic gliders; under $\sigma = 10275364$ a different regime of rule-domains forms (green and blue, under 2% Rule 110). The operator does not merely stir rule space: it drives it toward particular, computationally distinguished rules. Genotype colours are the legacy engine’s own.

2008) specialised to (1): our operator is a Boolean network whose signed interaction digraph is σ with every arc negative, a negation-cycle of length ℓ has sign $(-1)^\ell$, and an isolated negative cycle admits no fixed point while a positive (even) one admits exactly two. We contribute a machine-checked proof of the specialised statement in Lean 4 / Mathlib.⁶

The closed form

The number of $\sigma \in S_{2n}$ with all cycles even is $((2n-1)!!)^2$, with exponential generating function $\sum_n ((2n-1)!!)^2 x^{2n} / (2n)! = 1/\sqrt{1-x^2}$ (the “set of even cycles” construction). For S_8 this is $105^2 = 11,025$ of 40,320, or 27.34%. The count is classical: it is OEIS Foundation Inc. (2011) and appears as Stanley (1999, Problem 5.34(c)) and in Riordan (1968). We claim it only as the enumeration of our operator family, not as a new identity.

Fixed bytes number 2^k

An all-even σ with k cycles therefore has exactly 2^k fixed bytes, and any σ with an odd cycle has none.⁷ Each even cycle admits exactly two alternating colourings, chosen independently across cycles, whence 2^k ; the zero case is Theorem 1. This too is machine-checked.⁸ Summed against these weights the family is consistent: $\sum 2^k$ over all-even σ equals $(2n)! = 8! = 40,320$, the total number of fixed bytes across the family.

Every Fixed Point Is Balanced

Every fixed point has $\lambda = 1/2$

Every fixed point of every operator in the family has exactly four one-bits—equivalently, Langton’s activity parameter is pinned at $\lambda = 4/8 = 1/2$.⁹

⁶ `DarkTower/Patterns/Propagator.lean`, theorem `hasAlternatingColouring_iff_cycleType_even`; `lake build clean`, no `sorry` or `axiom`.

⁷ Supplement 2, Corollary 1: The 2^k count.

⁸ `alternatingColouring_card_eq_two_pow_cycleType_card`.

⁹ Supplement 2, Theorem 2: Every fixed point is balanced.

By Theorem 1 each cycle has even length $2m$; an alternating colouring assigns such a cycle exactly m ones; summing over cycles gives $(\sum L)/2 = 8/2 = 4$.¹⁰ Exhaustive enumeration confirms it: across all 40,320 operators and all their fixed points, every one has Hamming weight four, with zero exceptions.¹¹ The statement is machine-checked for Fin 8.¹²

We are deliberate about the standing of this result. Langton’s critical value was obtained empirically, as a statistic over rule space (Langton, 1990), and its status as an order parameter was contested: Mitchell et al. (1993) refuted Packard (1988)’s specific genetic-algorithm clustering claim, not the definition of λ , which survived as one axis among others such as Wuensche’s Z (Wuensche, 1999). There is a folklore intuition for why $1/2$ is special — binomial sampling variance $\lambda(1 - \lambda)$ is maximal there — and it is not ours; and it has been noted that λ alone does not fix a rule’s dynamical class, distinct classes coexisting at a single value (Sakai & Kanno, 2002). What Theorem 2 adds is narrower and firmer than any of these: under the complementation in (1) the value $1/2$ is *forced* as an algebraic matter, because any alternation-under-complement colouring is balanced by parity. We claim balance, exactly; any connection of that balance to dynamical behaviour we treat as interpretation, not as part of the theorem.

The cast: which rules survive, and how many

Under an all-even σ , the fixed points of P_σ —the rules it maps to themselves—number 2^k , where k is the number of cycles of σ (*The Classification*): each even cycle admits two alternating colourings, chosen independently across cycles. In the spatial runs these are the rules onto which the field is observed to settle. Every one has $\lambda = 1/2$ (Theorem 2), so the observed terminal population is a cast of *distinct* rules—distinct truth tables—sharing a *single* activity.

¹⁰ Supplement 2, Definition 3: Langton’s activity λ .

¹¹ Supplement 2, Observation 2: Balance is a truth-table property.

¹² `alternatingColouring_true_card_eq_four`: the true-cells and false-cells are put in bijection by σ , forcing each to number four.

The smallest nontrivial case of the count is a single cycle, $k = 1$: two survivors. The 2015 paper’s Figure 8 exhibited a two-rule residual of its own: a random field of tens of distinct rules dies off to an alternating population of just two. That resemblance is not an instance of this count. The writing in that example is non-bijective (*The 2015 Example*); the count above is a statement about permutations; and that writing admits no fixed rule whatever, its residual being a hunt on a single unsheddable position rather than a pair of fixed points. Two-rule residuals arise here for a different reason than they do there, and we keep the two apart.

That die-off curve—the count of distinct rules present over time—is a natural population-level companion to λ . Where λ is a scalar assigned to a single rule, this curve is a property of the whole field’s trajectory; and where λ is uninformative here—every fixed point sits at $1/2$ —the curve still varies, because it measures the population rather than the point. We use it cautiously, as an observable rather than an established order parameter, to tell the rotations whose fields stay diverse from those whose fields collapse (*Fixed Points Are Not Attractors*, Figure 6).

Diversity of the observed population and singularity of its activity thus coexist; the metric measures we tried do not separate the two (*Limits of the Tested Distance and Complexity Measures*).

A dead genotype can carry a live phenotype

The balanced shell is a definite object: exactly $\binom{8}{4} = 70$ rules carry four active outputs, and by Theorem 2 the absorbing set of every operator in the family lies wholly inside it. Collapse is therefore not merely a loss of diversity but a fall onto this particular 70-rule balanced shell at $\lambda = 1/2$.

Balance is necessary for a genotype to freeze and silent about what the frozen rule then does. Run each of the 70 as an ordinary elementary CA and they split: 32 are phenotypically live—sustained change under iteration, the additive rules 90, 105, 150 and the class-IV rule 54 among them—while the rest fall quiescent. The two halves share a

single λ ; this is, inside our family and by construction rather than by survey, the coexistence of dynamical classes at one activity value that has long been urged against λ as an order parameter (Sakai & Kanno, 2002). A field can thus die in its genotype while the surviving rule keeps its phenotype in motion.

Figure 4 is such a field, built to specification. To seat a collapse on a chosen live rule R it suffices to pair R 's four active positions with its four inactive ones as an involution: four transpositions, an all-even operator that holds R among its fixed points (Theorem 1). Whether the field actually settles onto that fixed rule is a separate, dynamical question (as *Fixed Points Are Not Attractors* stresses); here it does. Carried out for Rule 54 this yields $\sigma = [23015476]$; run from a random field it drives the genotype to near-uniformity—two rules survive—while the phenotype never settles. The field has died onto Rule 105, another live tenant of the same shell, whose additive dynamics keep the black-and-white panel full of structure. Genotypic death and phenotypic death are not the same event.

One caveat marks the open edge of this. An involution fixes not its target alone but the whole shell compatible with its pairing, and which tenant a random field actually reaches is set by the basins, not by the construction: we aimed at 54 and the dynamics preferred 105. Existence is settled—live survivors are constructible—but *steering* a field onto a named rule is not, and is the natural next question.

The Census

The parity dichotomy in the spatial dynamics

Theorem 1–2 concern fixed bytes; the spatial dynamics adds stochastic elementary writes and, in the census, neighbour blending. Figure 5 shows why fixed-point existence is not itself a dynamical verdict: without blending, all-even rotate+2 settles within the run while all-even rotate+1 remains diverse; the odd-cycle examples, which lack fixed bytes, also remain diverse. The full census below (*The full census, tabulated by class*) makes this quantitative over all 40,320 operators (Table 1); the dynamical content is the *death-rate difference* between the classes, not the class sizes.

A prospective test: reduced rotate+2

To guard against reading structure into a completed enumeration, we fixed a prediction before running the case. The reduced rotate+2 operators act as a single even cycle on a subset of positions and leave the remaining positions as one-cycles; by that leftover they are *not* all-even, so by Theorem 1 they possess no fixed point and should fall on the no-fixed-points side of the dichotomy—no rule for the field to settle onto, and correspondingly less extinction. They do (Figure 5, right): they stay diverse rather than dying off. Absence of fixed points forbids settling onto a single rule; it does not forbid short-period behaviour, and we claim only the observed persistence of diversity, not literal aperiodicity.

The full census, tabulated by class

The trend becomes an exhaustive finite-run observation once every operator is tabulated.¹³ We ran all 40,320 permutation-propagators and grouped them by cycle type (Table 1).¹⁴

Fixed Points Are Not Attractors

Possessing a fixed point (Theorem 1) is not the same as reaching one, and P_σ does not by itself move a rule toward $\lambda = 1/2$.¹⁵ Because P_σ complements every bit, $\lambda(P_\sigma g) = 1 - \lambda(g)$: the operator *reflects* activity across $1/2$, exactly preserving the distance $|\lambda - 1/2|$ rather than contracting toward it. And because P_σ is a bijection on bytes, a generic rule lies on a periodic orbit and simply cycles—its fixed points are fixed, but they are not attractors and have no basins. The only rules pinned at $\lambda = 1/2$ are the fixed points themselves. Attraction in the experiments therefore belongs to the stochastic elementary-write dynamics, with or without neighbour coupling, rather than to iteration of

¹³ Supplement 2, Definition 4: Census protocol and aliveness.

¹⁴ Supplement 1, Box 1: The full permutation census.

¹⁵ Supplement 2, Worked example 4: Reading the rotation survey.

Table 1

Genotype-alive fraction by cycle type under the census protocol, over all 40,320 permutations. Any odd cycle forbids a fixed byte (Theorem 1); empirically, every such operator is alive. The all-even types split, with collapse becoming more frequent as the number of cycles k increases.

cycle type	count	genotype-alive
any type with an odd cycle (17 types)	29,295	100.0%
[8]	5,040	82.9%
[2 6]	3,360	70.0%
[4 4]	1,260	66.7%
[2 2 4]	1,260	49.2%
[2 2 2 2]	105	25.7%

the simultaneous map P_σ .¹⁶

Constructive control by temporal schedules

Everything so far has held one operator fixed for the length of a run. Relaxing that—alternating the elementary spatial updates of two operators from step to step, rather than composing them into a single map—turns out to be constructive rather than merely destructive, and it is the first place in the paper where a schedule does something neither of its constituents can.

Braiding two operators that each collapse alone can sustain the field.¹⁷ Figure 8 shows the case: offset +2 (two 4-cycles) and offset +4 (four 2-cycles) are each genotypically dead run on their own, and alive when braided. The schedule is also a usable control rather than a switch. Mixing a sustaining operator with a collapsing one in fixed proportion over

¹⁶ Supplement 1, Box 2: The rotation survey.

¹⁷ Supplement 2, Definition 5: Temporal braid. Supplement 1, Box 3: A braid of two collapsing operators can sustain the field.

a periodic twelve-step schedule moves late-window rule diversity smoothly with the mixing fraction (Figure 9), which is what later lets us treat diversity as a dial to be turned rather than a property a run happens to have.¹⁸

Each operator negates every bit, so under composition the flips cancel in pairs: a product of two is a *pure* coordinate permutation with no negation left.¹⁹

Exploratory searches over products of two and three simultaneous propagators found no additional structural regime.²⁰ Temporal braiding is a different operation: it alternates stochastic elementary writes inside the spatially coupled dynamics rather than replacing them by one composed map.²¹

Algebraic composition is therefore closed in the sense of Proposition 1. Temporal scheduling is different: it retains each intermediate coupled state, which is why the braids of *Constructive control by temporal schedules* can produce dynamics absent from either held operator. A second constructive mechanism pairs a propagator with a step of a different kind.

Limits of the Tested Distance and Complexity Measures

The classification is a conjugacy invariant of σ —its cycle type—while the complexity and compression measures we tried read a single rule’s bits, or one space–time trajectory, without explicitly encoding σ . In our tests none of them recovered the parity classification. A population count reads a collapsed-but-maximally-selected field as nearly dead. A frozen periodic genotype scores a lower statistical complexity C_μ (Crutchfield & Young, 1989) than rule 110. And gzip rates a self-similar rule as noise. We report this as a limitation of the summaries we tested rather than as a finding: it is a negative result about those particular measures, not an impossibility theorem for trajectory-based inference, and

¹⁸ Supplement 1, Box 4: The mixing fraction is a reproducible diversity control.

¹⁹ Supplement 2, Proposition 1: Composition parity.

²⁰ Supplement 2, Definition 6: Interrupter.

²¹ Supplement 1, Box 5: The rotation split occurs with and without neighbour blending.

unlike the findings collected in Supplement 1 it rests on exploratory runs we did not preserve in reproducible form.

A Parameter-Space Test

The survey (Figure S3.1) exposes a sharp behavioural split within a family of provably fixed-point-bearing operators: only the single 8-cycles (offset $\pm 1, \pm 3$) sustain a high-diversity phase, while offset ± 2 and ± 4 collapse. A reader is entitled to ask whether that phase sits at a *critical point*—tuned between an ordered and a disordered phase, as CA complexity is often supposed to (Langton, 1990). This is the first of the two readings distinguished at the outset—an edge of chaos located in *parameter space*, at some value of a tunable control parameter—and the finite-size scan below is the test of it, the only place in this paper where that reading is put to the question. We then ask whether the sustained diversity is spatially organised into coexisting dynamical regimes.²²

Figure 10 collects the four diagnostics. The scan is over the propagator fraction q , from all blend at $q = 0$ to all propagator at $q = 1$, at lattice widths $L = 30$ to 240 with 32 seeds each; the point of varying L is that a genuine transition sharpens as the system grows while a crossover does not. Genotype activity fits a logistic in q at every size ($R^2 > 0.99$), but the crossover width does not shrink over these sizes ($w \sim L^{-0.04}$). The size-scaled run-to-run spread $L \text{Var}(a_G)$, used here as a susceptibility proxy, does peak inside the band and grows about 5.7-fold, but its maximum wanders—from $q = 0.75$ to 0.50 and then to the sampled boundary $q = 1$ for the two largest sizes—rather than settling. Finite-horizon collapse becomes rarer with size, and the Binder cumulant has no common crossing. Each diagnostic is individually consistent with a critical point; none of them locates one, and together they support a broad crossover instead.

The genotype field is also spatially organised. Tinting by activity resolves it into coexisting low- and high-activity domains whose boundaries trace fractal-like filamentary walls through spacetime; those walls carry elevated genotype churn, and at matched

²² Supplement 1, Box 6: No finite-size evidence of a critical point over the tested range.

activity the two sides hold measurably different rule populations, so they are not an artefact of the tint that draws them. We report that structure in Supplement 4, together with a measure of information transport along those boundaries that we built, validated against three nulls, and then withdrew when it proved to be an artefact of its own construction. Neither is a third edge, and nothing in the argument below depends on them.

Part II

Beyond the Permutation Core: Feedback and Causal Propagation

Part I's classified object and exhaustive census were the finite permutation core; its temporal braid was the boundary case, already replacing one held operator with a schedule. We now retain the same genotype–phenotype lattice and elementary rule semantics but drop the single-operator restriction. The objects in Part II are update architectures: time-dependent schedules, phenotype-conditioned writes, mutation and holding protocols, and conservative local transports. Some use writings from the 8^8 family as components, but a state-dependent architecture is not itself one fixed writing. Consequently, the results below neither classify nor measure the $8!$ family as a whole; they test how causal propagation changes when the genotype update can read the phenotype.

A Feedback-Coupled Construction

The next construction crosses that boundary by allowing the update to depend on the live phenotype. Figure 11 shows a representative run at $L = 240$: alternating phenotype domains branch, merge and reorganise for the length of the run rather than resolving to a homogeneous or short-period field.²³

A Causal Measure

A causal test also avoids a general problem with observational regions. Any statistic computed over a region drawn from the field it is computed on invites circularity, and no null defined only by moving that region can necessarily detect it. We therefore stop drawing regions: perturb one cell and ask where the effect arrives.

We run the phenotype-reading construction of *A Feedback-Coupled Construction* to

²³ Supplement 2, Definition 7: River. Supplement 1, Box 7: The river sustains structured phenotype.

$t^* = 60$, fork, flip a single phenotype bit at site x , and continue both branches to $T = 120$, recording which cells differ. The fork is exact: the construction draws one pseudo-random position per cell per step, so re-running from the same seed gives both branches an identical tape and every divergence is the causal consequence of the flip. The control is the matched feedback-off variant—same seed, tape, construction and initial state, with only the live phenotype-to-genotype edge cut. Figure 12 sweeps the flip over all 80 sites and records damage at a sequence of horizons.²⁴

Two readings of that result must be separated. A glider carries a fixed-width packet at constant velocity; a conductive medium disperses. Tracking the damage centroid distinguishes them, and the reference values must be measured rather than assumed—elementary rule 110, whose gliders support universal computation, does not give a fixed packet under a random single-site perturbation, because the perturbation creates and destroys gliders which then collide.²⁵

Figure 13 places every construction on that one axis. All points share the protocol of Supplement 1, Box 9, and the calibration reproduces rules 0, 204 and 90 exactly against it, so the bands drawn from the elementary rules carry over unchanged. The rows split the constructions by coupling rather than by provenance, and the split is clean. Nothing whose genotype update is blind to the phenotype clears the bottom of the complex band—not even the ungated transport controls, which are fully mobile and merely cannot see what they are moving through. Everything that reads the phenotype lands inside. The river appears in both rows, at 12.97 with its edge and 5.51 without, so the arrow between them is one construction moved by adding an edge rather than by changing a parameter.

The same sweep also gives the growth of the damage cone and the displacement of its centroid (Figure 14); we report those exponents descriptively and draw no inference

²⁴ Supplement 1, Box 8: Phenotype-to-genotype feedback roughly doubles causal reach.

²⁵ Supplement 1, Box 9: Causal reach on a scale calibrated against elementary rules. Supplement 1, Box 10: The order parameter is the strength of the phenotype-to-genotype coupling.

from them about gliders or particles.

The coupling as a dial

Treating that coupling as present or absent understates it. In both constructions that read the phenotype it is a quantity, and causal reach is a graded, monotone function of it.²⁶

For conservative transport the gain is the phenotype-gated swap rate; reach crosses rule 90 between rates 0.10 and 0.20 and saturates past 0.75. For the river it is the *currency* of the phenotype bits the genotype reads: each cell at each step reads the live phenotype with probability γ and a frozen reference otherwise, so γ is the fraction of the genotype's view that is causally current. Because the frozen field is itself a real phenotype, marginals and spatial structure are matched at every γ and only causal currency varies; because $\gamma = 1$ reproduces the live river exactly, the dial is anchored against a measurement made independently of it.

Two consequences bear on the reading of *A Causal Measure*. First, the $\gamma = 0$ endpoint is 1.28, not the 5.51 of the ablated control: that control freezes the phenotype the genotype reads but still presents a field which, in the perturbed branch, contains the flip. It cuts the *currency* of the read, not the read. The doubling reported for it is therefore the smaller part of a coupling effect that spans an order of magnitude.

Second, the two dials have opposite curvature (Figure 15). Transport is concave and buys most of its reach early; the river is convex, with 45% of its span in the last eighth of γ , so one stale read in eight costs 40% of the reach. Monotonicity in the coupling is thus common to both constructions while its shape is not, and we would not generalise the shape from two cases.

²⁶ Supplement 1, Box 11: Coupling is a dial, not a switch, and reach is monotone along it.

Diversity and conservative transport

Langton’s λ is silent on this family: the classification puts *every* fixed rule of *every* operator at $\lambda = 1/2$, so it cannot distinguish an operator whose field stays diverse from one that collapses. It cannot be rescued by computing it on the realized dynamics either.

Across the nine configurations of Figure S3.2 the empirical neighbourhood distribution is close to uniform—normalised visitation entropy 0.946 to 0.999, all eight neighbourhoods visited in every run—so λ read off the realized transitions with uniform weight (0.42 to 0.52) and the same quantity weighted by visitation (0.40 to 0.51) differ by 0.035 on average, and both sit near $1/2$ wherever the nominal value does. Nor is the silence peculiar to this family: of the elementary rules calibrating the scale above, rules 30, 54, 90, 150, 204 and 232 all have $\lambda = 1/2$ exactly, spanning frozen, ordered, complex and chaotic. What the balance theorem contributes is not a new failure of λ but an exact and provable instance of a known one. The sustained distinct-rule count can, and the causal measure above lets us ask the stronger question: does causal reach depend on diversity itself, or on the mechanism used to sustain it?

Our initial survey varied sustained diversity by seven mechanisms that share little: the bare operator family; braided writings; the fork time after seeding the genotype with a randomised *full* rule population; a continuous blend strength; random rule mutation; asynchronous refuges that hold a cell’s rule exactly with some probability; and spatially phase-shifted braid niches. A limiting case anchors the top of the range: a genotype held fixed at the full population has maximal diversity and no rewriting at all. The coordinate throughout is diversity *sustained through the response window*, not diversity at the instant of the fork—a field seeded with all 256 rules holds them for one step and collapses during the very window the damage is measured in.²⁷

Figure 16 shows why the apparent optimum was an artefact. The original

²⁷ Supplement 1, Box 12: Sustained diversity does not determine causal reach; conservative transport breaks the apparent interior optimum.

configurations trace a rise and fall against sustained diversity, but every point on the descending limb obtains its diversity either by replacement mutation or by holding rules still, so the descent measures those two mechanisms rather than diversity. The conservative runs, which hold the rule histogram exactly invariant while varying transport, separate into flat bands — one per transport rate — that span the whole diversity axis. Reading across a band shows the effect of diversity at fixed transport and finds almost none; reading between bands shows the effect of transport and finds an order of magnitude.

Two cautions bound this result. First, the conservative intervention is a local transposition, rather than the one-bit, one-site intervention of the original survey. Normalising by initial damage puts their amplification on a common scale, but the conservative curves are a falsification control, not a claim that their absolute heights rank all mechanisms. Second, phenotype-gated bijective exchange is one deliberately constructed transport law, tested at one lattice width and one response horizon. It establishes existence: fields can remain mobile and propagate damage at effective diversities well above 135 without mutation. It does not establish that transport rate is a universal order parameter.

The defensible conclusion is narrower and cleaner. Distinct-rule diversity describes population persistence in the Part I operator experiments, but it does not govern causal reach across the Part II architectures. The latter depends on how rewriting or transport is organised. Identifying a coordinate that captures that organisation across mechanisms is work for another paper; the classification of *The Classification* guarantees only that λ cannot discriminate among fixed points in the permutation core.

What this scale does and does not license needs saying plainly. It separates systems that propagate a perturbation from those that do not, and that separation is sharp: frozen rules return exactly zero. It is *not* a regime coordinate. Wolfram class III rules bracket the range at both ends—rule 90 at 8.00 and rule 30 at 37.50—with the class IV rules 54 and 110 between them, so a position on this axis does not by itself classify a system as ordered, complex or chaotic, and we make no such classification. What we claim is narrower and

causal: the live phenotype-to-genotype edge roughly doubles how far a single-cell perturbation reaches, on a measure that involves no region-drawing and therefore cannot fail in the way the withdrawn measure of the Supplement 4 analysis failed. It concerns the phenotype-coupled construction, not the bare operators; no comparable measurement of the family is attempted here.

What the coupling supplies

One measurement bounds how the convexity of *The coupling as a dial* should be read. Asking, per cell and step, whether the live phenotype context selects a *different* rule than a frozen context would: it does in 73.1% of cell-steps (and in 76.8% against no context), over four seeds at $L = 80$ for 120 steps. The construction's first-match runs over four candidates, so an uninformative re-selection would differ in about 75% of cases: at 73.1% a frozen context carries almost no information about the live answer. The read is determinative rather than a tiebreak, which is what a convex gain curve should lead one to expect, and it means there is no common case that a phenotype-blind rule could encode in its place. We draw no stronger conclusion from this than that: whether *any* blind construction could match the river is a question about a search we have not run, and Supplement 1, Box 8 reports only that sixteen hand-chosen ones do not.

Part III

Endogenous Coupling

In Part II the gain of the phenotype-to-genotype loop is a parameter we set. This part asks what happens when it is not: when the coupling is a function of the lattice's own local state rather than a dial turned from outside. The objects are still compositions of operators Part I classifies, so nothing new is introduced at the level of the substrate; what changes is who sets the gain.

An Endogenous Gain

A *conditional composition* applies one operator where a local predicate holds and another where it does not: $\text{switch}(c, A, B)$. Throughout this part A is conservative transport at unit rate, which sits in the chaotic band of the calibration of Section II at 25.38, and B is holding, which sits in the ordered band at 1.21. The predicate decides, cell by cell and step by step, which applies. Its *firing fraction* f —the proportion of opportunities at which A is selected—is measured from the run rather than prescribed.

Two disciplines make the comparison below meaningful rather than merely suggestive, and both are inherited from the gain dial of Section II. First, the transport stream advances once at every opportunity whether or not the gate fires, so a predicate that reads the phenotype cannot desynchronise the two branches of the damage fork and manufacture divergence that is not causal. Second, where the predicate consumes randomness of its own, those draws come from a separate stream re-seeded identically in both branches, so the gate never itself injects difference.

The first question is what a composition does when its condition carries no information about the lattice. A gate firing at a prescribed rate, blind to the phenotype, is well described across six rates spanning $f = 0.10$ to $f = 0.94$ by a line through the holding value,

$$\text{reach}(f) = 1.21 + 22.42 f, \tag{2}$$

with the unconstrained fit returning $R^2 = 0.993$. A blind conditional composition is therefore *exactly* a gain dial: its duty cycle plays the role γ plays in Section II, and reach is linear in it. This is worth stating plainly because it is the null against which everything else in this part is measured, and because it means the construction adds nothing on its own. Making the gain a function of *something* does not by itself buy anything; it matters what that something is.

A Reading Gate Exceeds the Dial

Now let the predicate read the phenotype. Write $\text{agree } k/m$ for the condition that at least k cells of the centred circular m -cell phenotype neighbourhood agree with the centre. Thirteen such gates were placed on the same scale, spanning widths $m \in \{3, 5, 7, 9\}$ and, at each width, strictness from bare majority to unanimity.

Every one of them exceeds the blind line of Equation 2 at its own measured firing fraction, and several do so by a wide margin. The largest departure is +22.8 cells; the widest permissive gate, $\text{agree } 5/9$, reaches 38.23 while firing on only 63% of opportunities—not merely above the blind line but above unit-rate transport’s own 25.38. Firing a chaotic operator less often, in the right places, spreads damage further than firing it always.

What predicts the size of the departure is the *width* of the neighbourhood the gate reads, not how hard the gate is to satisfy. Adding width to a quadratic-in- f model of departure raises R^2 from 0.84 to 0.96, at +1.47 cells per cell of width; adding strictness instead raises it only to 0.88. The design separates the two—the correlation between width and strictness across the thirteen gates is +0.01—whereas strictness is nearly collinear with the firing fraction itself at -0.91 , so its apparent effect is largely f under another name. The model-free version of the same statement is that within each of three groups of gates matched on f , departure increases monotonically with width; in the group near $f \approx 0.65$ the strictness ratio is almost constant at 0.56–0.60 while the width nearly doubles.

Currency, Not Correlation

That result admits a deflationary reading, and it is the reading its own mechanism suggests. A gate reading m cells has neighbouring decisions sharing $m - 1$ inputs, so width *is* the spatial correlation length of the firing pattern. The blind gate of Equation 2 draws an independent coin per cell and is therefore spatially uncorrelated. Correlated firing of a *transport* operator composes adjacent swaps into longer-range movement, so a departure that grows with width might be pure geometry, with the phenotype playing no part beyond

supplying a conveniently correlated field.

The control that separates the two possibilities holds the phenotype still. A *frozen* gate applies the identical agree k/m predicate to the phenotype as it stood at t^* : same predicate, same width, same spatial statistics, a real phenotype field, and therefore the same correlation structure in its firing. What it cannot do is track where the perturbation has since spread. Each branch of the damage fork keeps its own t^* snapshot, so both frozen fields contain the perturbation itself and only the live gate sees its consequences—the ablation of Section II applied to the gate rather than to the update.

Across seven pairs matched on (m, k) , the live gate exceeds its frozen counterpart in every case, by 14.5 cells on average. More tellingly, every frozen departure from the blind line is *negative*, between -0.04 and -4.18 . A gate with the same width and the same correlated firing, reading a field that is merely stale, sits at or slightly below the spatially uncorrelated baseline. Spatial correlation on its own buys nothing here; if anything it costs a little. The comparison is also conservative in the direction that matters: the frozen gate fires *more* often than its live counterpart in six of the seven pairs, so it is given more opportunities and still reaches far less.

The reading we take from this is that the gate is a targeting mechanism, and targeting works only on current information. Firing transport where the phenotype is *now* locally uniform lifts causal reach far above a blind gate at the same rate; firing it where the phenotype *was* uniform at t^* does not, though the two firing patterns share width, spatial statistics and rate. So the coordinate that governed in Part II governs here too, reached by a different route: an endogenous gain does not escape the causal currency of what the genotype reads, it re-derives it. A composition whose condition is blind reproduces the dial exactly; a composition whose condition reads the phenotype exceeds the dial precisely to the extent that what it reads is current.

Discussion

Specifying an Experiment Before Running It

A selection experiment reports trajectories. Nothing in a trajectory says whether the experiment was capable of exhibiting the effect it was built to detect, and a run that could not have succeeded produces the same shape of output as one that tried and failed. The apparatus described here exists to separate those two cases in advance rather than after the fact.

The target as a certificate

We state the target as an object that must be exhibited, not as a pattern to be recognised in a plot. A genome carries a rule at each locus together with a flag marking that locus as held; a held locus keeps its inherited rule for the whole evaluation. Writing dep for the dependence of a genome on within-evaluation rewriting and perf for its raw function, a completed assimilation claim is a path $g_0, \dots, g_n$ satisfying, simultaneously:

1. every step $g_i \rightarrow g_{i+1}$ is admitted by the declared genetic operators;
2. $\text{perf}(g_i) \geq \tau$ for every i , so the path never leaves the functional region;
3. $\text{dep}(g_{i+1}) \leq \text{dep}(g_i)$ at every step, and $\text{dep}(g_n) < \text{dep}(g_0)$ strictly;
4. g_n is static — every locus held; and
5. perf of g_n with every locus held still clears τ .

The last condition is the one most easily omitted. A population can reduce its measured dependence without its inherited field being able to carry the behaviour alone, and only an explicit held-rule evaluation distinguishes the two.

Selection, however, does not rank raw function. It ranks fitness $\text{fit}_c = \text{perf} - c \text{dep}$. A path can therefore satisfy all five conditions and still be untraversable, because a step that preserves function may lose fitness. Adding the requirement $\text{fit}_c(g_{i+1}) \geq \text{fit}_c(g_i)$ gives a

strictly stronger object, and makes a conflict explicit that is otherwise rediscovered empirically: a dependence-reducing step keeps its fitness exactly when

$$\text{perf}(g_i) - \text{perf}(g_{i+1}) \leq c \left(\text{dep}(g_i) - \text{dep}(g_{i+1}) \right), \quad (3)$$

that is, when the function a hold costs is no more than the cost it saves. If the inequality fails at every accessible dependence-reducing step, no traversable path exists at any length, and (3) identifies c as the parameter that would dissolve the obstruction.

Conditions checkable before the run

Two consequences are checkable without running anything. First, if dependence takes a single value across the reachable genomes, the strict-decrease condition cannot be met and no certificate of any length exists; a constant axis is therefore a disqualifying property of the design, not a null result. Second, and less obviously, an axis on which the score is non-constant may still be untraversable: a profile that is zero at every level but one has two distinct values and a single transition, and a hill-climber cannot cross a lone cliff. Counting the adjacent transitions that carry gradient, rather than counting distinct values, is what distinguishes a navigable axis from a spike.

This distinction is not hypothetical here. Under the gain parameterisation used in *An Endogenous Gain*, sampled at eight uniform levels, the band score is zero at seven of them and positive at the eighth. The design is disqualified by the second criterion while passing the first, and re-resolving the parameter where the gain curve is convex is what makes the axis traversable — a change to the experiment rather than to the check.

Invariants the implementation must satisfy

The specification above constrains the target. A second family of conditions constrains the apparatus, and each is stated as an executable predicate tested against the specific failure that motivated it, since a check written from the description of an invariant rather than from its counterexample can pass the defect it was introduced to catch.

Tape alignment. Both branches of a perturbation fork must consume identical random

draws whatever their gates decide. A gate that reads the phenotype fires at different loci in the two branches, so a closed gate that skips a draw desynchronises the tapes and the divergence that follows is an artefact of the generator rather than a causal effect. The discipline is to draw unconditionally and then decide how to use the draw; the check must also confirm the branches actually differed, since equal draw counts are trivially equal when they do not.

Layering fidelity. Any extension must reproduce the measurement it extends when its additional machinery is in the neutral position. Determinism does not imply fidelity: an extension can reproduce byte-for-byte and still report a different value than the construction it claims to generalise.

Linked inheritance. Every heritable component must travel together under recombination. Transferring a rule without its held flag places it under an unrelated hold pattern, so the operator no longer tests the mechanism it was introduced for.

Operator reachability. Every value of every gene must be attainable under mutation. A value the operator cannot reach is present in the representation and absent from the experiment.

Empirical null. Any claim of the form “rarer than chance” requires a no-selection control running the identical lifecycle. An analytical baseline must match the lifecycle actually implemented: with elitist retention, in which survivors are not re-mutated, the mutation-only expectation for a symmetric flip rate μ after n generations is $\frac{1}{2}(1 - (1 - \mu)^n)$ rather than the familiar $\frac{1}{2}(1 - (1 - 2\mu)^n)$.

Treatment separation. Selection ranks scores, so a term that shifts every genome equally cannot act. Arms differing in such a term will coincide, and their coincidence is a diagnostic of an inert treatment rather than a robust result.

Two of these deserve emphasis because they invert a natural reading. The empirical

null is not a formality: in the runs reported here, dependence falls and the held fraction rises under no selection at all, purely because holds accumulate by mutation. Declining dependence is therefore not evidence of assimilation, and a treatment must be compared against that trajectory rather than against zero. Treatment separation is the complementary case: identical arms indicate that the parameter never reached selection.

Diagnosing a negative result

When a run produces no certificate, the specification does not say why, and the three possible reasons call for different remedies. Holding each locus of a high-performing plastic genome at each admissible rule separates them. If no rule preserves function at a locus, the plastic computation there is not representable by any single static rule. If some rule preserves function but the inherited allele does not, the representation suffices and the difficulty is that mutation cannot prepare the allele before it is fixed. If the inherited allele itself preserves function and improves fitness, an assimilation step was available and the search did not take it. Only the first is a statement about the substrate.

The distinction is decidable cheaply. Testing each locus at its current allele costs one evaluation per locus rather than one per locus and rule, and settles the third case outright when it fires.

What the apparatus does not establish

These conditions are necessary, not sufficient. They constrain what an experiment can claim; they do not make its measurements correct, and no invariant here would detect a mis-specified dynamics that satisfies all of them. They are also specific to the construction studied in Part III: the certificate assumes a genome with per-locus holding and a scalar cost, and neither the conditions nor the diagnostic transfers unchanged to a different architecture. Finally, the calibration used by the score must share dynamics with the constructions placed on it; boundary conditions that differ between a calibration and the objects it calibrates can reorder the reference rules themselves, and where that holds the resulting band assignments are not comparable.

Limits

The Part I theorems are exact for the Boolean permutation core; its dynamical claims are narrower and empirical. The census uses three seeds per operator, the rotation survey two, and the finite-size scan 32 per parameter cell, all under the stated radius-one spatial protocol. These finite-run results do not turn fixed-point balance into a claim about attraction. The substrate itself is not new (*The Object*), and λ is one heuristic among several (Wuensche, 1999); what is exact is only that the fixed-point structure of Equation (1) forces every fixed rule in the core to have $\lambda = 1/2$.

Part II has different limits because it has a different object. Its central causal comparison concerns one feedback-coupled construction rather than either writing family, one lattice width, and six seeds. The descriptive exponents are fitted over a range where damage width reaches only a tenth of the lattice, so finite-size effects are not excluded. The architecture survey and conservative-transport control compare mechanisms rather than exhaustively sampling 8^8 writings, and their fixed sample sizes are stated with each result. They identify feedback and transport as controls in the tested constructions, not as universal order parameters. Illustrative diagrams use pinned representative seeds. All reported computations are reproducible from the accompanying package.

Supplement 4 carries the separate spatial-domain analysis and the information measure withdrawn after its mask and measure proved circular. Its claims and failures do not enter either Part I's classification or Part II's causal comparison.

Conclusion

The paper has studied three related objects, and its conclusions must keep them separate. Part I concerns the finite permutation core. Its static results are exact: a permutation-propagator possesses fixed rules exactly when all cycles are even; an all-even permutation with k cycles has 2^k fixed rules; and every such rule is balanced, with $\lambda = 1/2$. Its dynamical results are finite-run observations over that core. The exhaustive census shows that all odd-cycle operators remain genotype-alive under the stated protocol while

every all-even cycle type contains both sustaining and collapsing members. The rotation survey and finite-size scan sharpen the distinction: fixed-point existence does not determine attraction, and the tested propagator-fraction scan gives a broad crossover rather than evidence of a critical point.

Part II retains the two-layer lattice but changes the object from an individual writing to an update architecture. The river construction composes a permutation-propagator with a phenotype-reading step; the wider comparison adds temporal schedules, mutation, holding, niches, and conservative local transport. These are not a census of the 8^8 writings, and the results do not retroactively characterise the $8!$ core. They establish a narrower causal claim. Against a matched feedback-off control, phenotype-to-genotype feedback roughly doubles the reach of a single-cell perturbation. When the strength of that feedback is varied, reach changes monotonically, while a control matched in rule mobility but blind to the phenotype remains in the low-reach band. Conservative transport further shows that sustained genotype diversity is not itself the governing coordinate: high diversity can coexist with propagation, and transport changes reach far more than diversity does.

The connection between the parts is therefore motivational rather than deductive. Part I proves that the classical λ coordinate cannot discriminate among fixed points in the permutation core and finds no critical point in the tested parameter scan. Part II asks what a causal measurement can reveal once the architecture itself is allowed to change. Its answer is feedback coupling for the constructions tested here, not a theorem about all permutations or all writings.

This admits a reading the substrate makes unusually clean. The gain γ interpolates between a genotype that reads only an ancestral phenotype and one that reads the currently acquired state, with the frozen field a real phenotype so that marginal statistics and spatial structure are matched at every setting; only causal currency varies. What the dial varies is thus how legible acquired state is to the heritable layer—the Lamarckian

question, posed as a continuum rather than a dichotomy. Its discriminating power is unusual in this family: across λ , propagator fraction, sustained rule diversity, mutation rate, refuge probability, niche width and rule mobility, causal reach never leaves the ordered band, and blend strength alone reaches its edge. The substrate carries no fitness, so this concerns what the heritable layer can see and not what is selected for.

Part III changes the object once more, from an architecture whose gain we set to one that computes its own. Its results are narrower than Part I's and of the same kind as Part II's: finite-run causal measurements over a chosen family of predicates, not a characterisation of conditional compositions in general. What they establish is a negative and a positive that belong together. A composition whose condition is blind to the phenotype reproduces the Part II dial exactly, linear in its duty cycle—so conditional composition is not by itself a new mechanism. A composition whose condition reads the phenotype exceeds that dial, by margins that grow with the width of the neighbourhood read; but an otherwise identical gate reading a phenotype frozen at t^* does not exceed it at all, and in every tested pair falls slightly below. Width is the spatial correlation length of the firing pattern, so this is the comparison that matters: correlated firing alone buys nothing, and what the reading gate exploits is that what it reads is current. The coordinate is therefore the same one Part II identifies, recovered from a construction that shares none of Part II's dials. That convergence is the strongest connection the paper draws between its parts, and it is still a convergence of measurements rather than a theorem. Supplement 4 records the separate spatial-domain analysis and the withdrawn boundary statistic; neither is a premise of this two-part argument.

Part III is narrower still. Its measurements cover one pair of constituents—unit-rate transport against holding—and one predicate family, agreement within a centred circular neighbourhood. Thirteen conditioned gates at four widths support the width gradient; seven matched pairs support the frozen control. Each row is sixteen seeds at one lattice width, so the between-seed spread is real and reported rather than smoothed. Two limits

deserve naming. The frozen gate’s firing fraction does not match its live counterpart exactly, differing by up to 0.04; the discrepancy runs in the direction that weakens the result, since the frozen gate mostly fires more often and still reaches less, but it is not zero. And while the frozen control shows that current information is *necessary* for the departure, it does not decompose the width gradient: how much of that gradient is currency and how much is the geometry of correlated firing is not resolved here. A blind gate with matched correlation length, rather than matched rate alone, would separate them and has not been run.

Future work. Part III closes the first of the two extensions the gain result suggested and sharpens the second. The gain is no longer only a parameter we set: a predicate reading the lattice sets it, and reach responds to whether what the predicate reads is current. What that leaves open is the step from a gate to a regulator. The conditions tested here are fixed families chosen by us; nothing yet adjusts its own condition in response to how it is doing. The 2014 ancestor of this substrate already contains the relevant construction—a switch that fires only where the local context carries no new information—so a system that tunes its own gate is buildable on this lattice rather than hypothetical, and the question is whether such a system settles anywhere in particular on the calibrated scale.

The second extension is unchanged and remains the more interesting. As it stands the substrate instantiates the Lamarckian arm alone: acquired state writes into the heritable layer, and both the dial of Part II and the gates of Part III vary the strength or the aim of that write-back. The Baldwin effect requires something this substrate lacks—a selective filter, so that plasticity alters what selection sees and the heritable layer follows without direct transcription. Adding selection over a population of these architectures would supply that ingredient, and because both arms would then be expressible in one family they could be placed on the single measure used throughout this paper: the reach of damage-spreading, calibrated against rules of undisputed regime. Whether direct

write-back and selection-mediated assimilation arrive at the same place on that scale, and by what route, is a question the family is now equipped to ask.

Data and code availability. Every numerical result in this paper is regenerated from source rather than deposited: the generators are seeded, so a single documented command rebuilds the fields underlying the tables, figures, and reported summary statistics from an empty data directory, and the analysis scripts consume no input that command does not produce. That command was run twice from an empty data directory and all 307 generated artefacts were byte-identical across the two runs; the 298 of them that also exist as published artefacts match those as well. The code and those commands are publicly available at <https://github.com/tothedarktowercame/mmca-clj> (commit 7e04a6de6afe35de041b0ac4bb52a5cea25fe8d4)—a standalone, dependency-light Clojure port of the MetaCA dynamics whose `README` gives the complete reproduction command and commands for individual figure groups. The classification and balance theorems are formally verified in Lean 4 with Mathlib at <https://github.com/tothedarktowercame/mathlib4>, branch `darktower` (commit bc34d8cec27216aae1d3cd511a9950ab1bce3575), in `DarkTower/Patterns/Propagator.lean`: the cycle-parity criterion as `hasAlternatingColouring_iff_cycleType_even` and the $\lambda = 1/2$ balance as `alternatingColouring_true_card_eq_four`. The same file also contains executable regressions for both hand calculations, including the displayed-position versus numerical-neighbourhood indexing contrast.

Generative-AI disclosure. Anthropic Claude (Opus 4.8, Opus 5, and Fable 5) and OpenAI Codex (version 5.6-sol) coding agents (Anthropic, 2026a, 2026b; OpenAI, 2026) assisted in an iterative, human-supervised workflow with manuscript editing, software implementation, test construction, and adversarial checks. Exchanges with Claude Fable 5 contributed to the initial identification of the operator family reported here. The author reviewed the resulting prose and code, reran the reported computations and formal

checks, and accepts full responsibility for the manuscript. Product information was accessed in July 2026; per-source access dates are given in the bibliography.

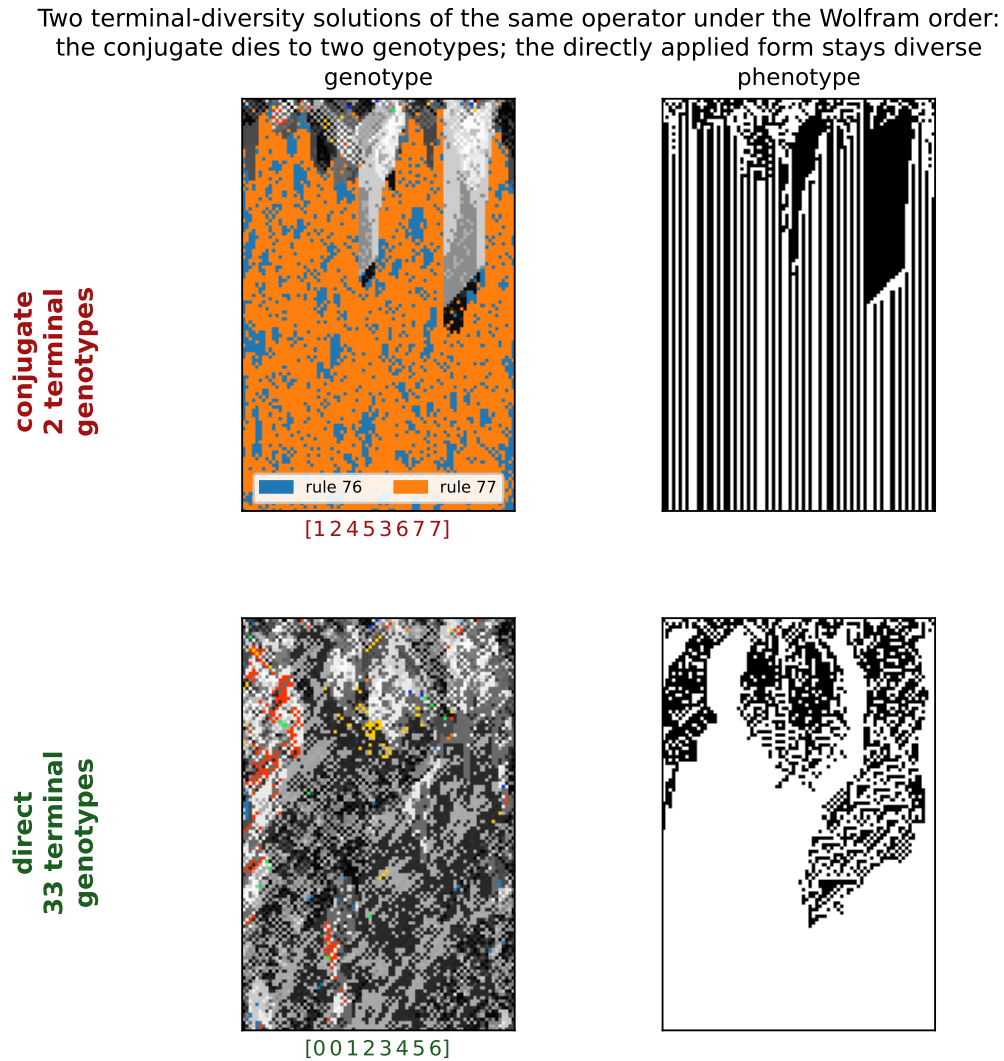
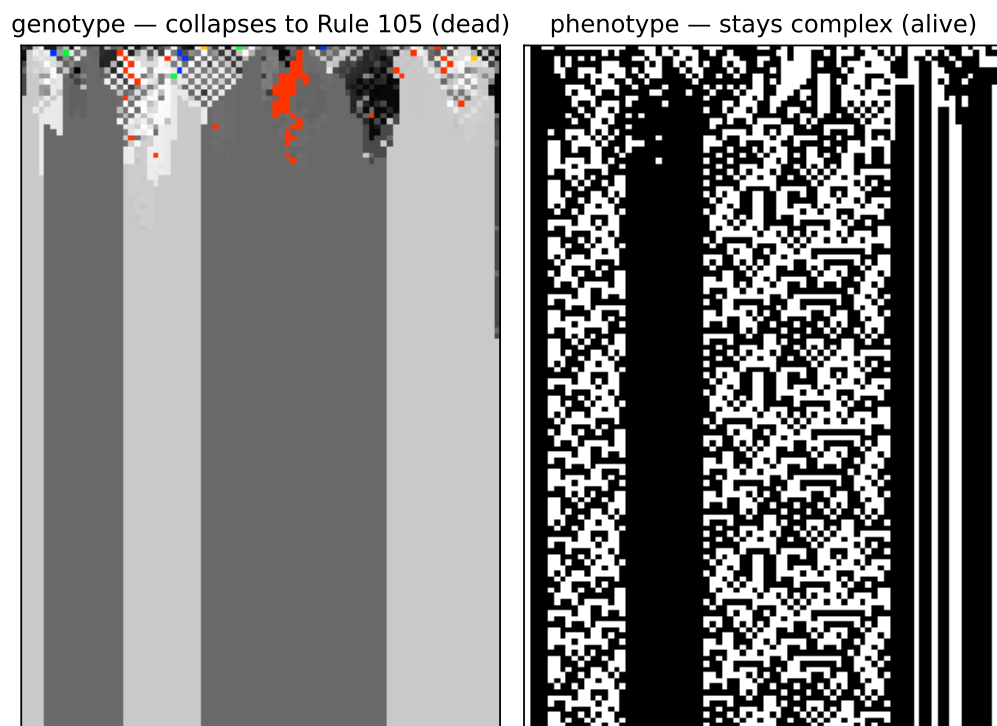


Figure 3

Two terminal-diversity solutions of one historical operator. The historical write [00123456] has two forms under the standard Wolfram order. Its conjugate [12453677] (top) reproduces the two-rule behaviour: the genotype collapses to exactly two terminal rules—76 and 77, adjacent, differing in a single bit—regardless of seed, while the phenotype stays regular. Applied directly (bottom), the same operator keeps a diverse genotype (~30 terminal rules). Two ends of genotypic aliveness from one historical write.

**Figure 4**

A genotypically dead, phenotypically live field. The all-even operator $\sigma = [23015476]$, built to fix the balanced shell containing Rule 105, collapses the genotype (left, 256-colour) to near-uniformity within a few dozen rows—two rules survive—yet the phenotype (right) never settles: the field has died onto Rule 105, an additive $\lambda = 1/2$ rule whose dynamics keep it complex. Seed 1, $W = 80$, $T = 120$.

CYCLE PARITY DICHOTOMY: a Boolean fixed point exists iff every cycle is even

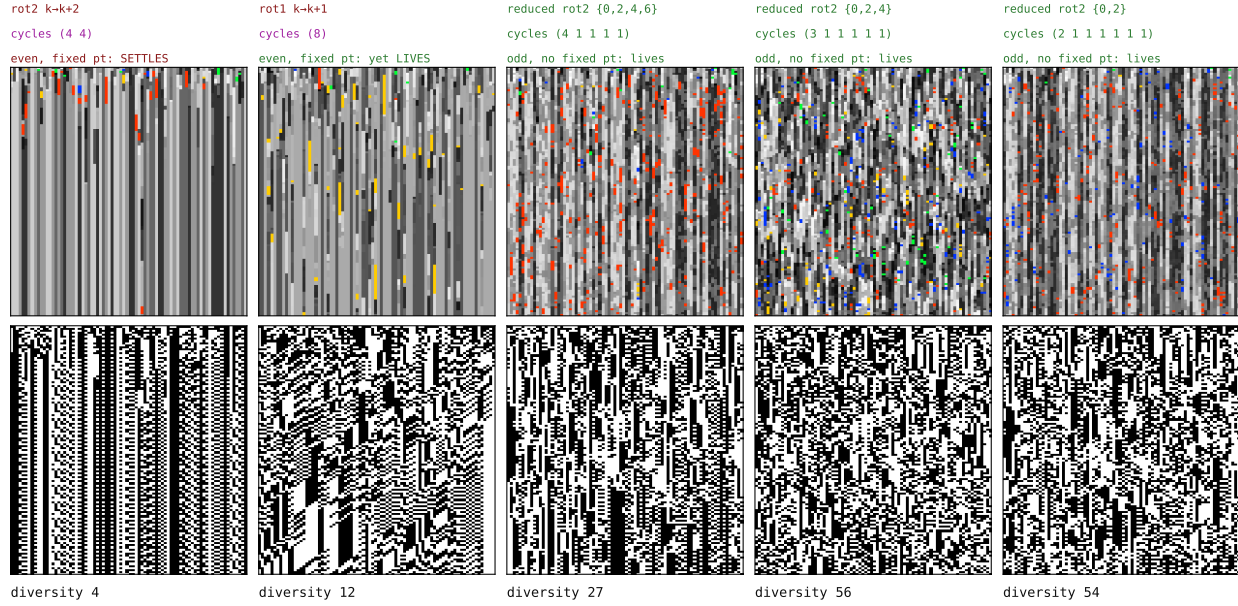


Figure 5

The parity dichotomy, by example (the census of all 40,320 permutations is summarised in the text). Five operators as genotype (top) and phenotype (bottom) over time. Left: both all-even operators possess fixed bytes (Theorem 1), but only rotate+2 settles within the run, reaching its $2^k = 2^2$ fixed rules; the single-8-cycle rotate+1 remains active at diversity 12. Right: the odd-cycle operators (reduced rotate+2, acting on a subset so the remaining positions are odd one-cycles) possess no fixed byte and stay diverse in these runs. All five use stochastic elementary writes without neighbour blending. The figure therefore illustrates the theorem's fixed-byte classification while also showing that all-even parity makes settling possible, not inevitable on a finite horizon.

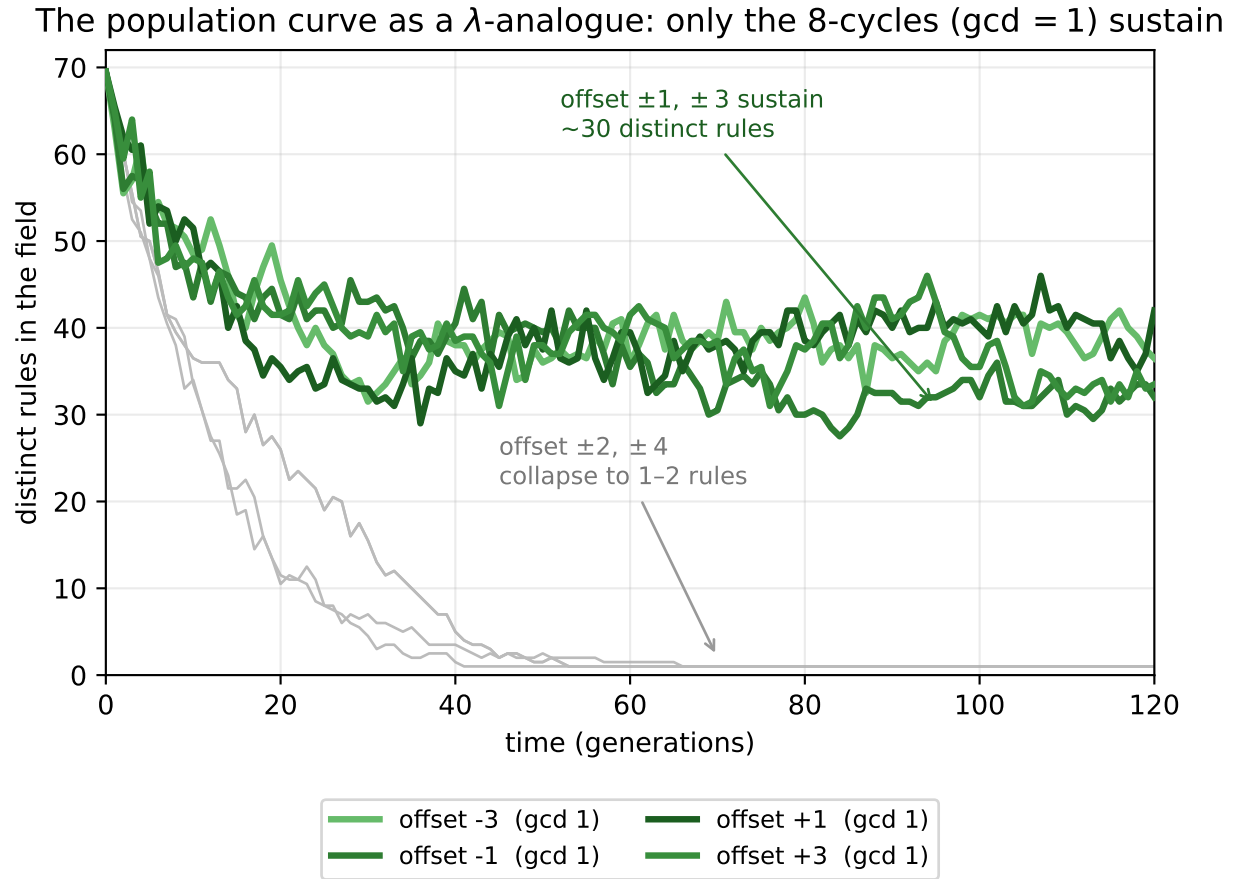


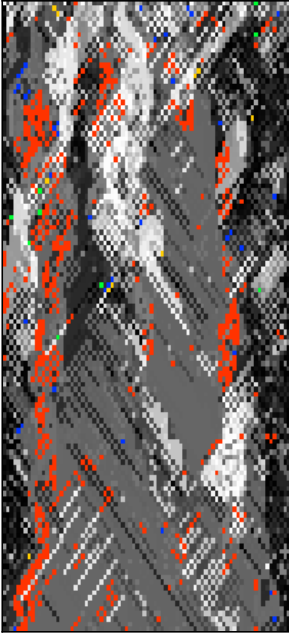
Figure 6

A population observable: distinct rules in the field over time, for the wrap rotations (two seeds each). All begin at ≈ 70 ; the four 8-cycles, $\gcd(\text{offset}, 8) = 1$ (offset ± 1 and ± 3 , green), keep a large diverse population, while offset ± 2 and ± 4 (grey) collapse to one or two rules. This makes quantitative, on the runs shown, the difference visible by eye in Figure S3.1. Figure 5 gives the corresponding without-blend comparison for offsets $+1$ and $+2$: the former remains active at lower diversity over that run, while the latter settles onto its four fixed rules.

Same cycle type (two 4-cycles), opposite fate: cycle structure does not determine aliveness

[6 7 0 2 1 4 3 5] — **two 4-cycles**

sustains (live / live)
genotype

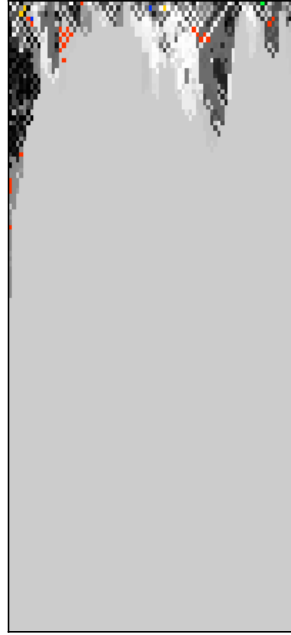


phenotype



[2 3 4 5 6 7 0 1] = **offset +2** — **two 4-cycles**

collapses (dead / dead)
genotype



phenotype



Figure 7

Cycle structure does not determine aliveness. Two operators of identical cycle type—both two 4-cycles—with opposite fate. Left, [6 7 0 2 1 4 3 5] sustains a live genotype and phenotype; it is the standard-Wolfram-order form of the historical offset+2 operator (its conjugate). Right, the Wolfram-order rotation offset+2 = [2 3 4 5 6 7 0 1] has the same two-4-cycle structure yet collapses to a dead genotype and a frozen phenotype. Same cycle type, opposite outcome: the specific permutation, not its cycle type alone, decides. A full census of the 40320 operators confirms the two-4-cycle class splits between sustaining and collapsing.

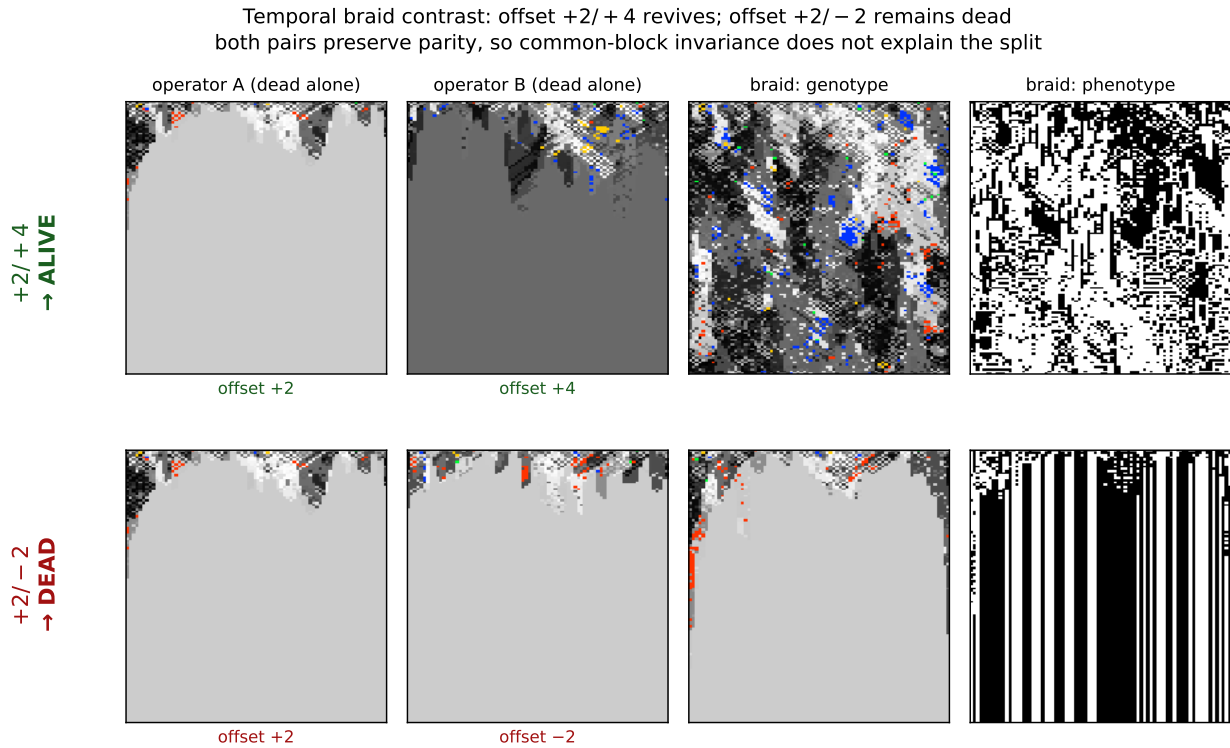


Figure 8

Temporal braiding can sustain aliveness. Two operators that each collapse alone are braided by alternating their elementary spatial updates. Top: offset+2 (two 4-cycles) and offset+4 (four 2-cycles) are each genotypically dead, yet their braid is alive (genotype change-rate 0.76, vs. 0 for either alone). Bottom: offset+2 braided with offset−2 stays dead. Both tested pairs preserve parity, so the contrast is not explained by the existence of that common invariant partition. The result concerns time-dependent coupled dynamics, not composition of the simultaneous maps P_σ .

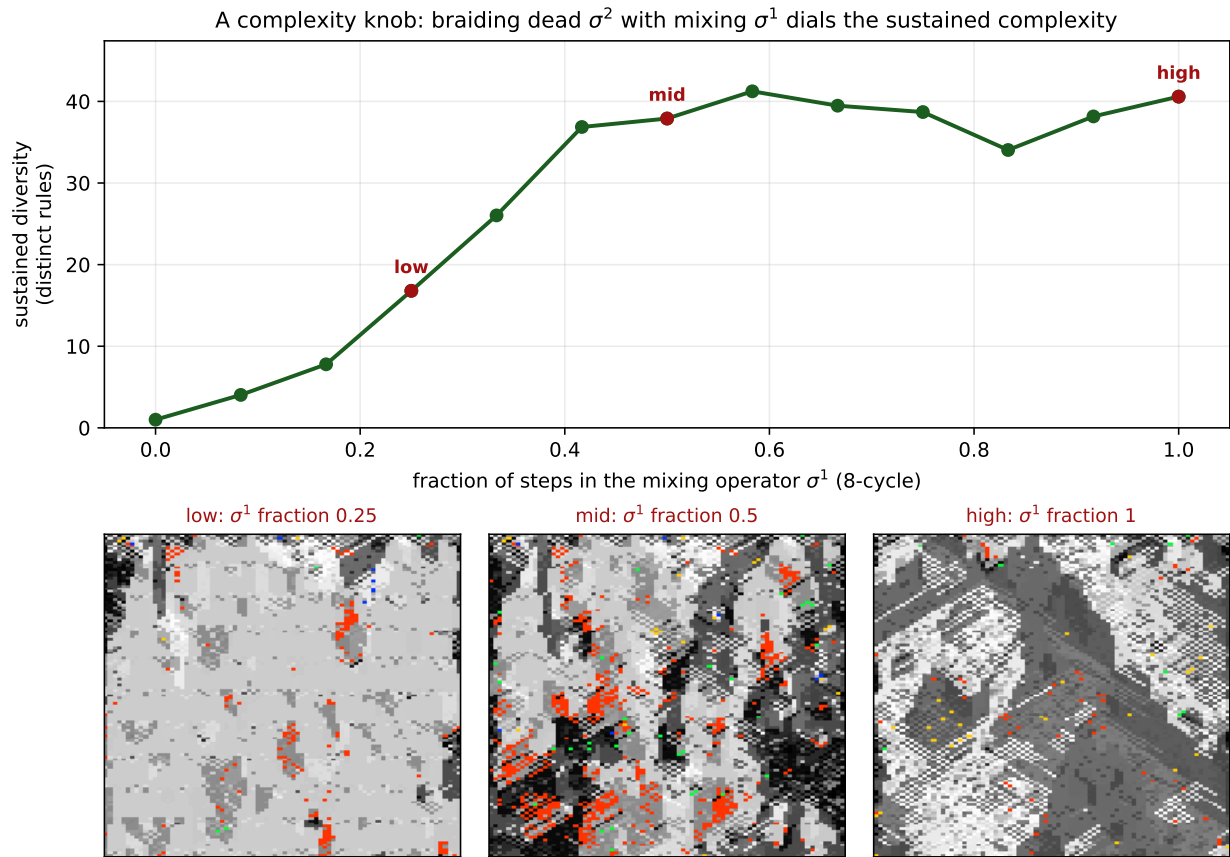


Figure 9

*A **diversity-control schedule**. Periodic 12-step schedules mix the sustaining offset+1 operator σ^1 with the collapsing offset+2 operator σ^2 . Mean late-window distinct-rule diversity (five seeds) rises from 1.0 with no σ^1 steps to a high-diversity plateau once its fraction reaches roughly 0.4. The three fields show fractions 0.25, 0.50, and 1.0 at the same representative seed.*

offset+1 feedforward scan across system size L (row width in cells, 30–240; 32 seeds): a smooth crossover with no critical point, and a susceptibility that peaks in the band but drifts and barely grows—metastability, not criticality (bottom: one realization across q , $L = 240$)

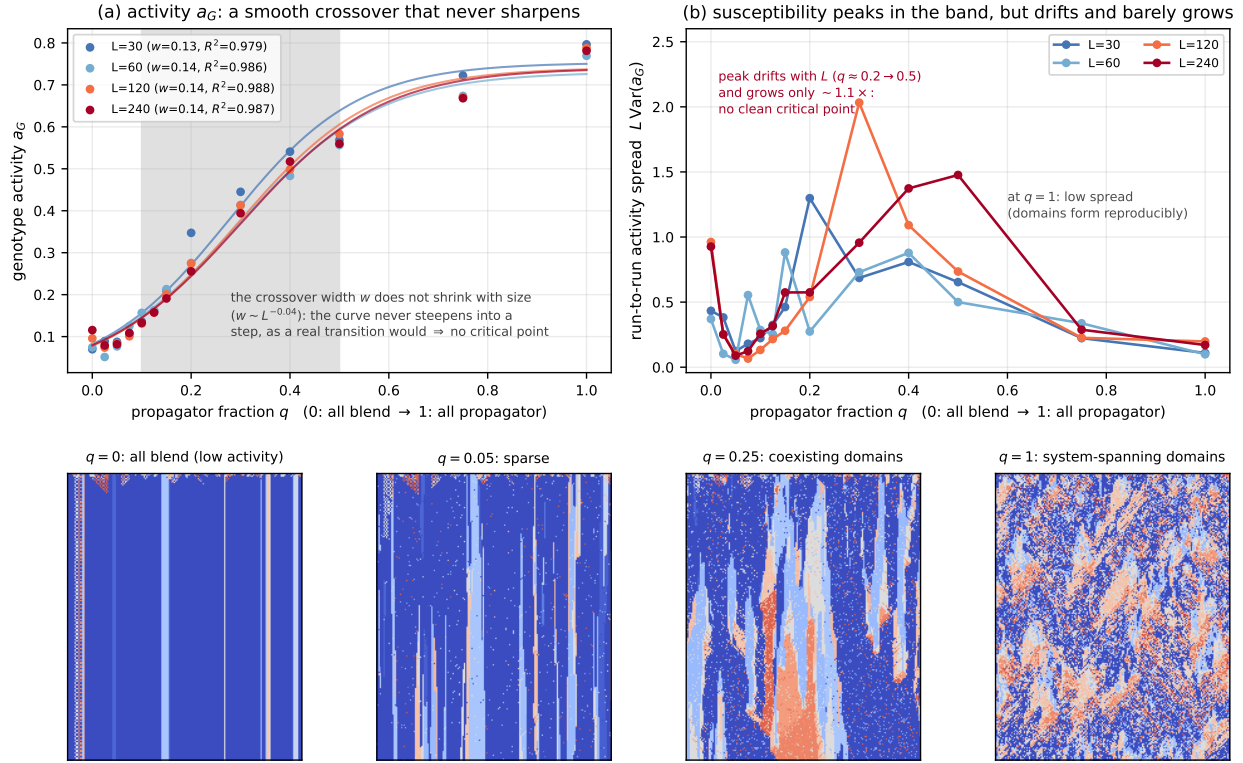


Figure 10

Finite-size scan across the propagator fraction. Offset +1 feedforward,

$L = 30\text{--}240$, 32 seeds per size. (a) genotype activity with logistic fits; (b) size-scaled spread as a susceptibility proxy; (c) finite-horizon collapse frequency; (d) the Binder cumulant.

Bottom: one paired realisation at $L = 240$ tinted by isolated-rule activity.

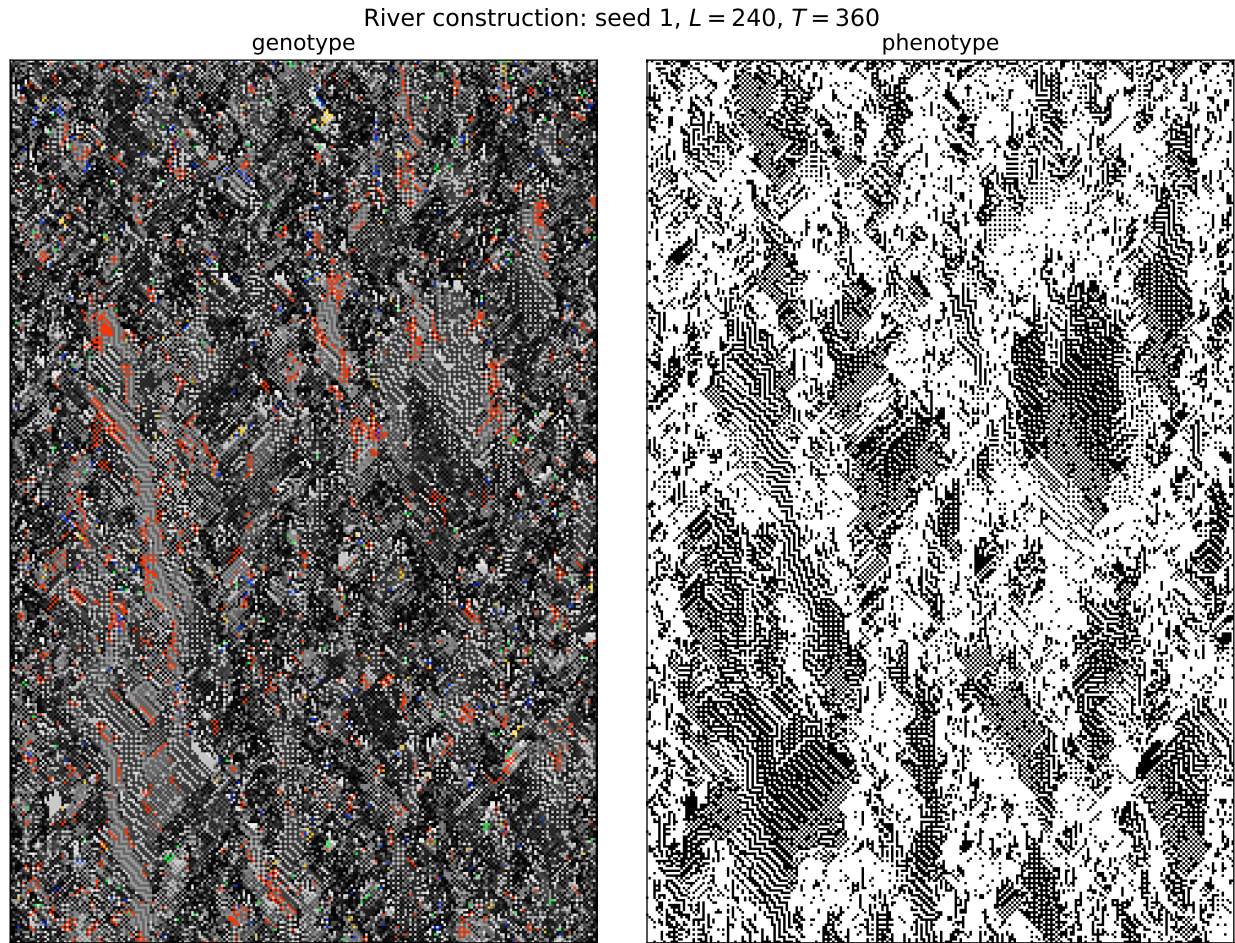


Figure 11

A larger river spacetime plate. One pinned representative run (seed 1) at $L = 240$, $T = 360$, with genotype at left and phenotype at right. The offset-+2 propagator is composed with a phenotype-reading, four-candidate, first-match template construction (fallback: the all-zero rule). Alternating phenotype domains branch, merge, and reorganise throughout the run rather than resolving to a homogeneous or short-period field. Genotype is coloured by rule as in Figures 2–Figure S3.1: greyscale by byte value, with well-known ECAs highlighted and red denoting Rule 110. Where the bare rotations collapse or cycle (Figure S3.1), this composition sustains structured phenotype for the length of the observed run.

A single phenotype-bit flip spreads farther in the river than in the feedback-cut control
(same seed, same RNG tape, same construction; single-seed sweep over all 80 sites, $L=80$)

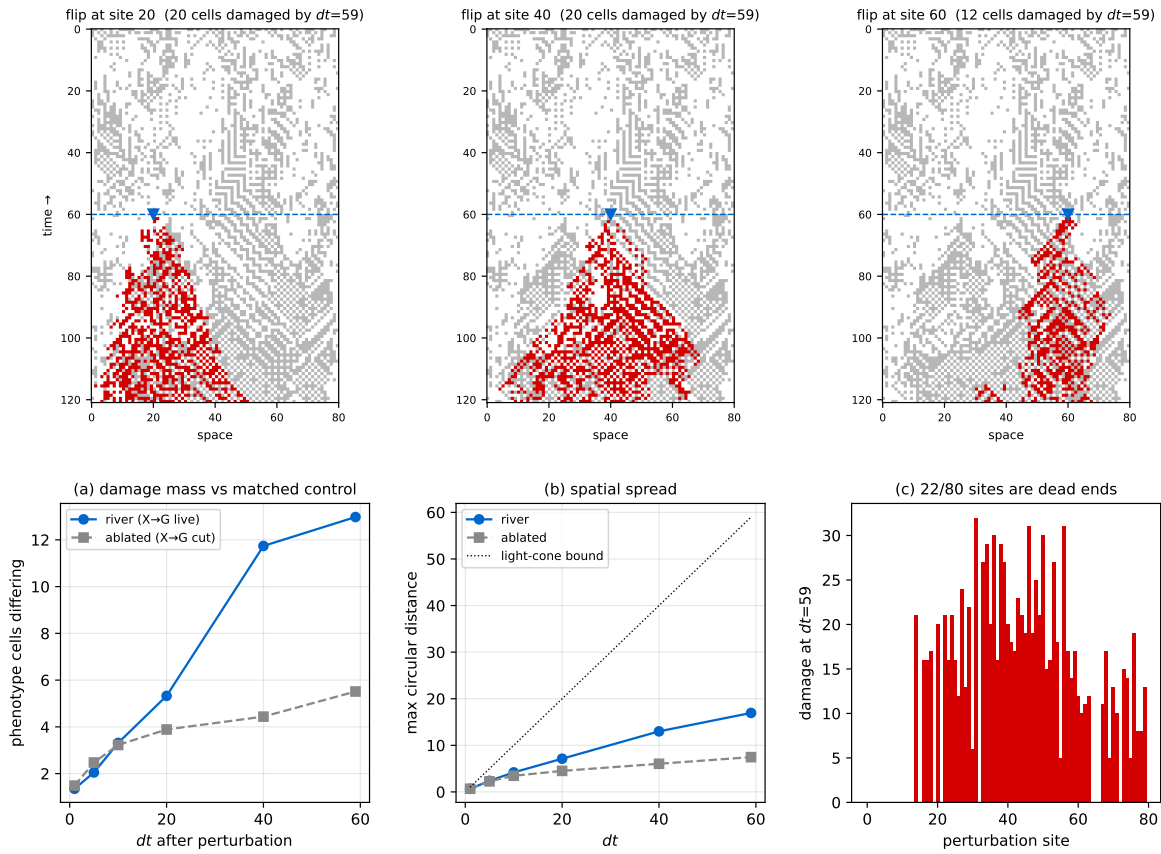


Figure 12

Damage from a single phenotype flip. Live construction against its matched control, $L = 80$, $t^ = 60$, swept over all 80 sites; mass and spread at a sequence of horizons dt .*

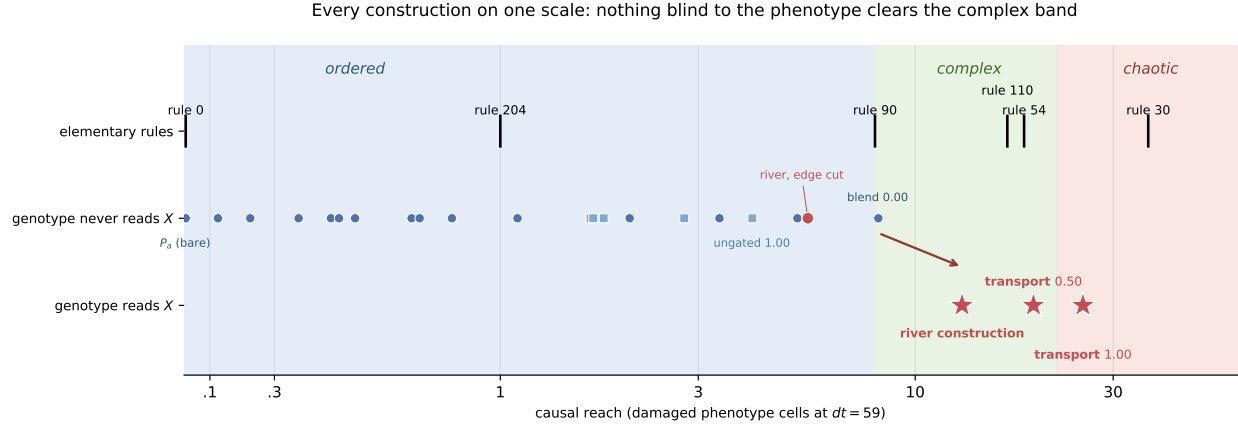


Figure 13

Every construction on one scale. Causal reach under the protocol of Supplement 1, Box 9, with regime bands drawn from the elementary rules. Rows divide constructions by whether the genotype update reads the phenotype.

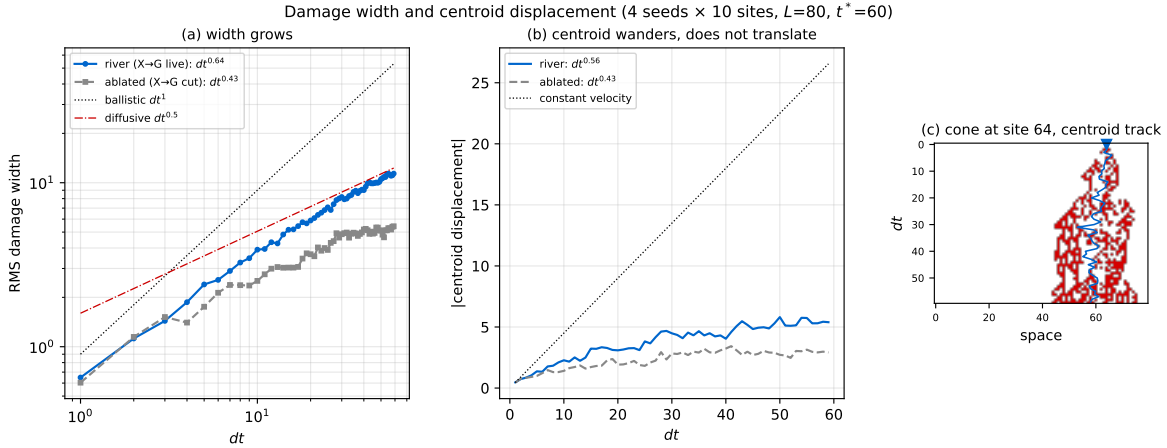


Figure 14

Damage width and centroid displacement. (a) damage width against dt , log-log, with ballistic and diffusive references; (b) centroid displacement; (c) a single cone with the centroid track overlaid. The packet disperses while its centroid stays near the perturbation site. We do not read this as excluding particle-like transport: under the same protocol rule 110, whose gliders support universal computation, gives $dt^{0.68}$ and $dt^{0.73}$, so a random single-site perturbation does not produce a fixed-width translating packet even where gliders certainly exist.

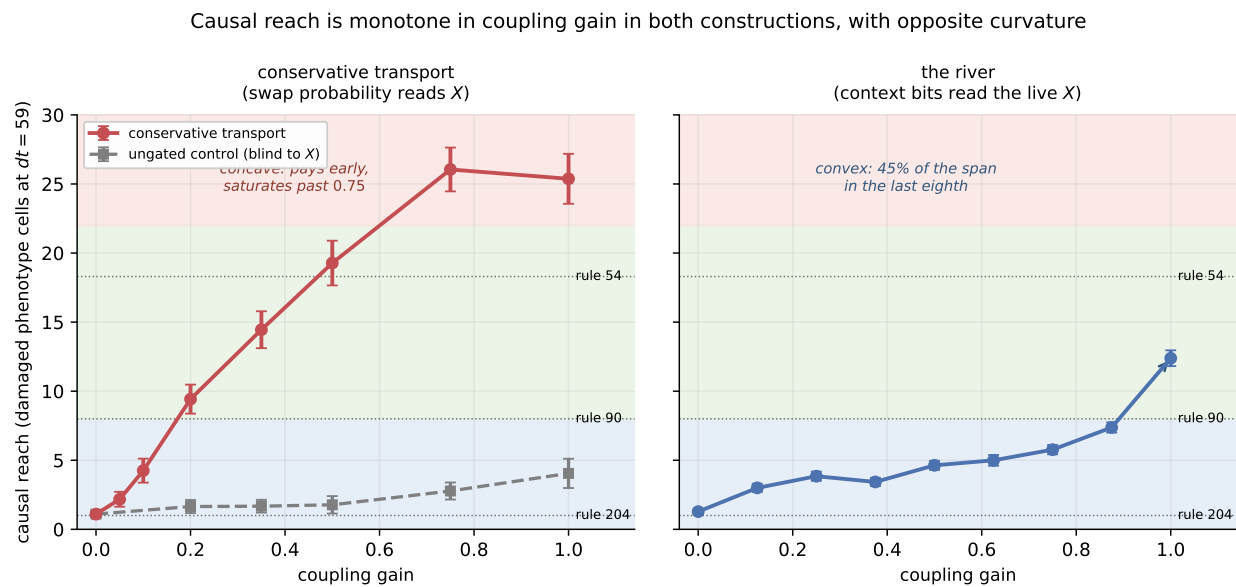


Figure 15

***Two couplings, dialled.** Causal reach against coupling gain for conservative transport (left, with an ungated control at matched mean swap probability) and for the river (right). Error bars are standard errors.*

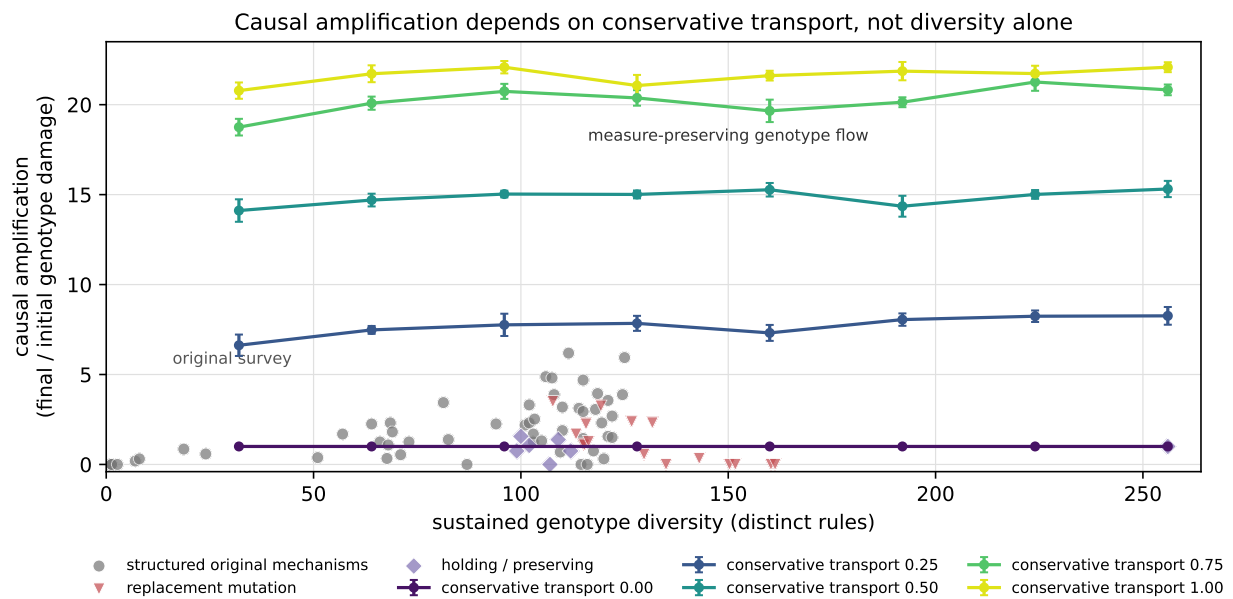


Figure 16

Causal reach against sustained genotype diversity. The original 76 configurations, grouped by mechanism, with the conservative-transport runs overlaid as one band per transport rate.

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